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BENGALURU, INDIA

WHERE THE WORLD COMES TO LEARN

TRADITIONAL MEDICINE AND HEALTH MANAGEMENT- BBT755C

Dr.Ashalatha

Department of Biotechnology



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Overveiw

Acharya stands as a beacon of excellence in higher education, boasting a legacy of academic distinction since its establishment in 1990. We offer a transformative educational experience, fostering holistic development, nurturing innovation and providing world-class facilities to ensure an enriching journey for our students.

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11 Institutions, Infinite Possibilities

We provide 100+ programs across 50 academic streams.



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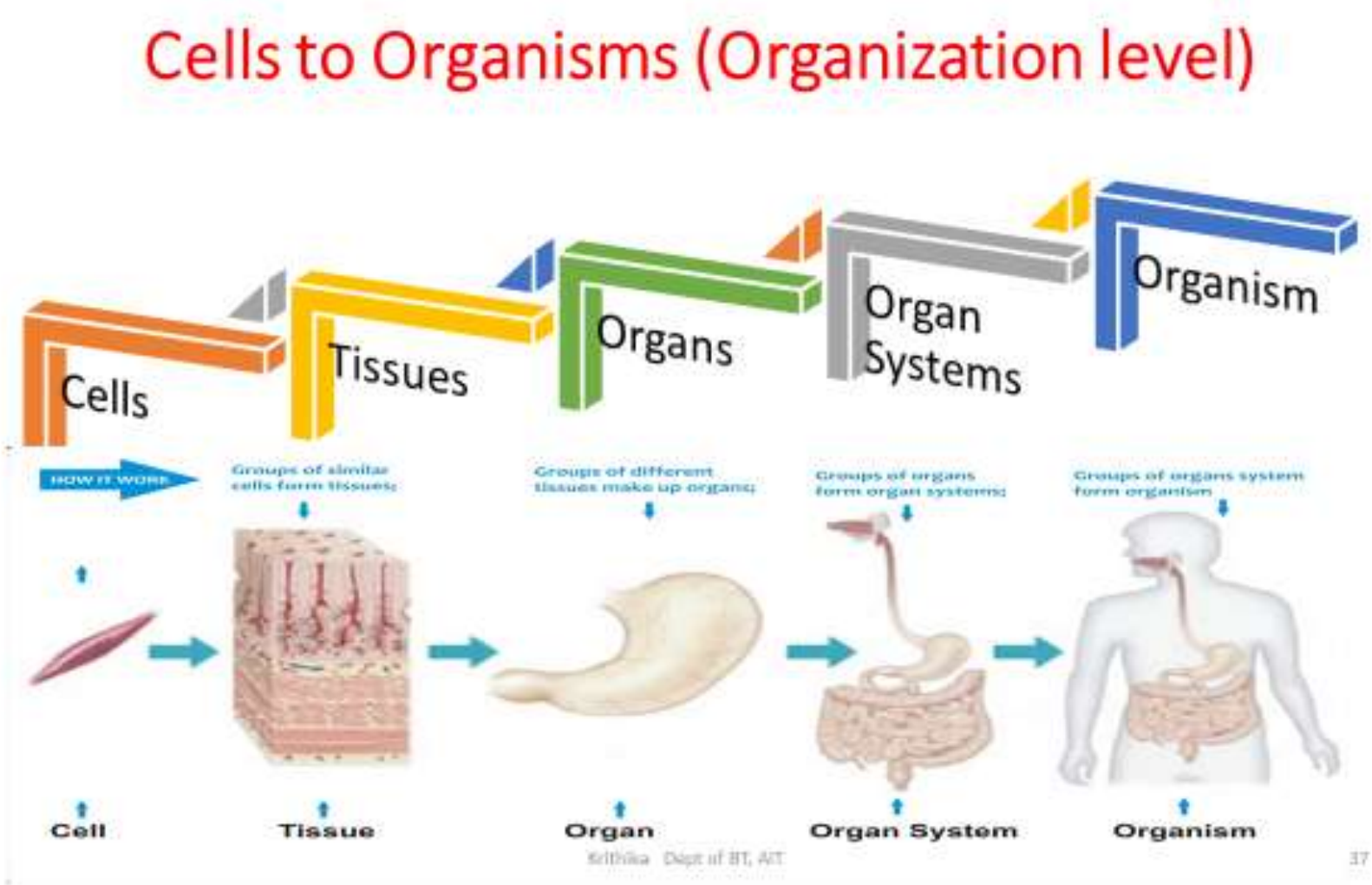
MODULE – 1 (8 Hours) HUMAN SCIENCES AND NUTRITION

- Brief introduction of - structure and functions of the cells, tissues and biomolecules, Metabolism and control of carbohydrates, lipids and proteins.
- Brief introduction of Structure and functions of the human organ system.

Brief Introduction to Cells, Tissues, Biomolecules

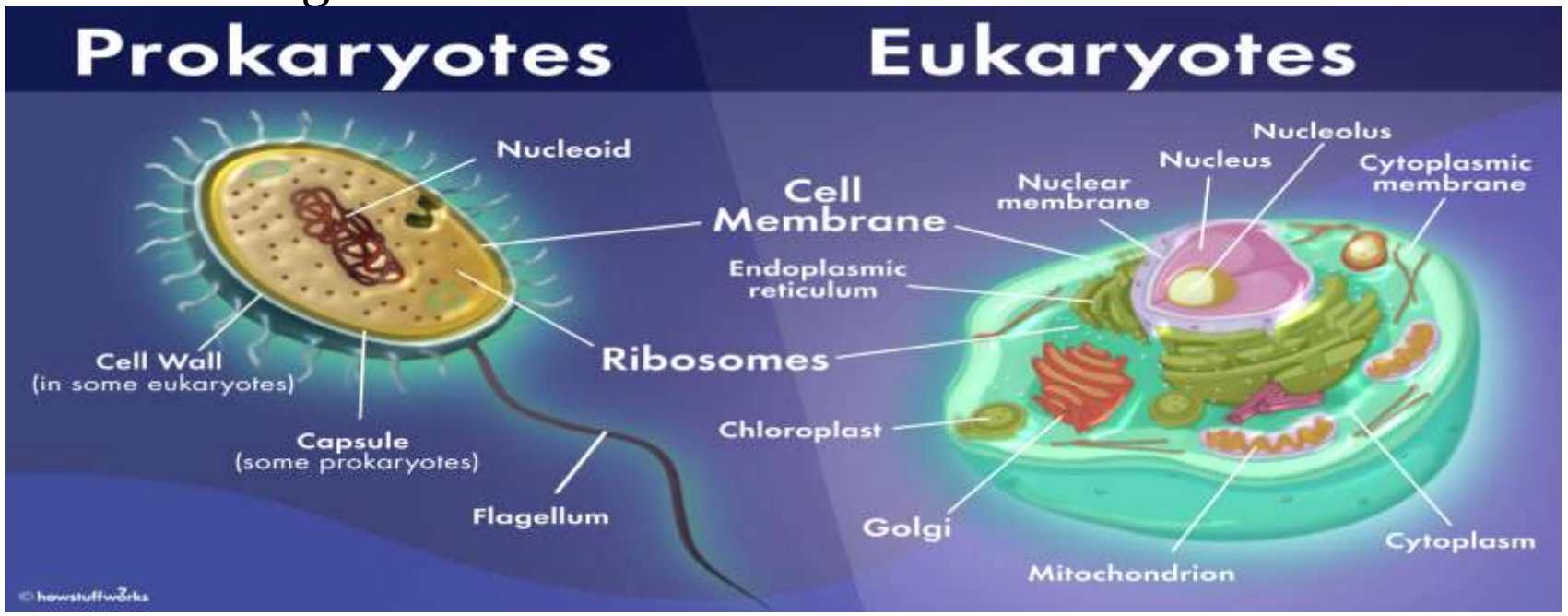
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CELL – *The Basic Unit of Life*

- **Structure of a Cell**
- Cells are the smallest structural and functional units of all living organisms. They exist in two forms:
- **Prokaryotic Cells** (bacteria, archaea) → simple, no true nucleus, no membrane-bound organelles.
- **Eukaryotic Cells** (plants, animals, fungi, protists) → complex, with nucleus and organelles.



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Parts of a typical eukaryotic cell:

- **Cell membrane (Plasma membrane):**
  - Thin, flexible boundary made of lipids and proteins.
  - Semi-permeable → controls movement of substances in & out.
- **Cytoplasm:**
  - Gel-like fluid inside the cell.
  - Contains organelles where metabolic activities occur.
- **Nucleus:**
  - Control center of the cell.
  - Surrounded by nuclear membrane, contains DNA (genetic material).
  - Functions in cell division and protein synthesis.
- **Cell Organelles:**
  - **Mitochondria** → "Powerhouse of cell", produces energy (ATP).
  - **Endoplasmic Reticulum (ER)**:
    - Rough ER → protein synthesis (with ribosomes).
    - Smooth ER → lipid synthesis and detoxification.
  - **Ribosomes** → protein factories (present freely or on ER).
  - **Golgi Apparatus** → modifies, stores, and packages proteins.
  - **Lysosomes** → contain enzymes for digestion ("suicidal bags").
  - **Vacuoles** → storage sacs for water, food, and waste.
  - **Chloroplasts** (plants only) → photosynthesis, contain chlorophyll.
  - **Cell wall** (plants, fungi, bacteria) → rigid protective covering made of cellulose.

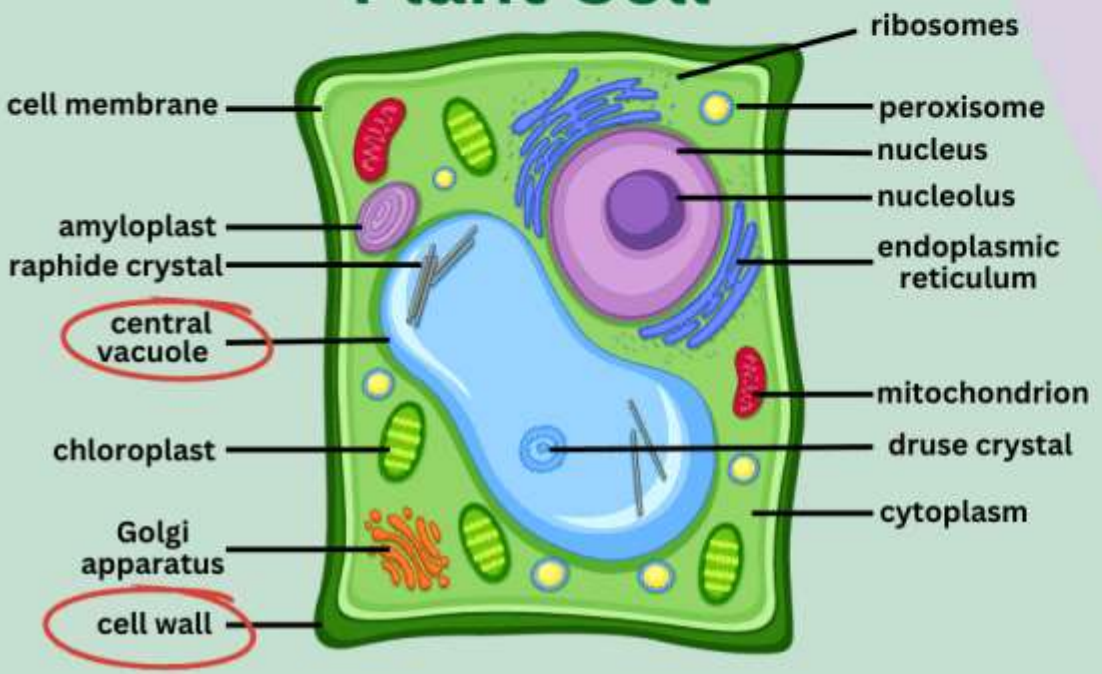
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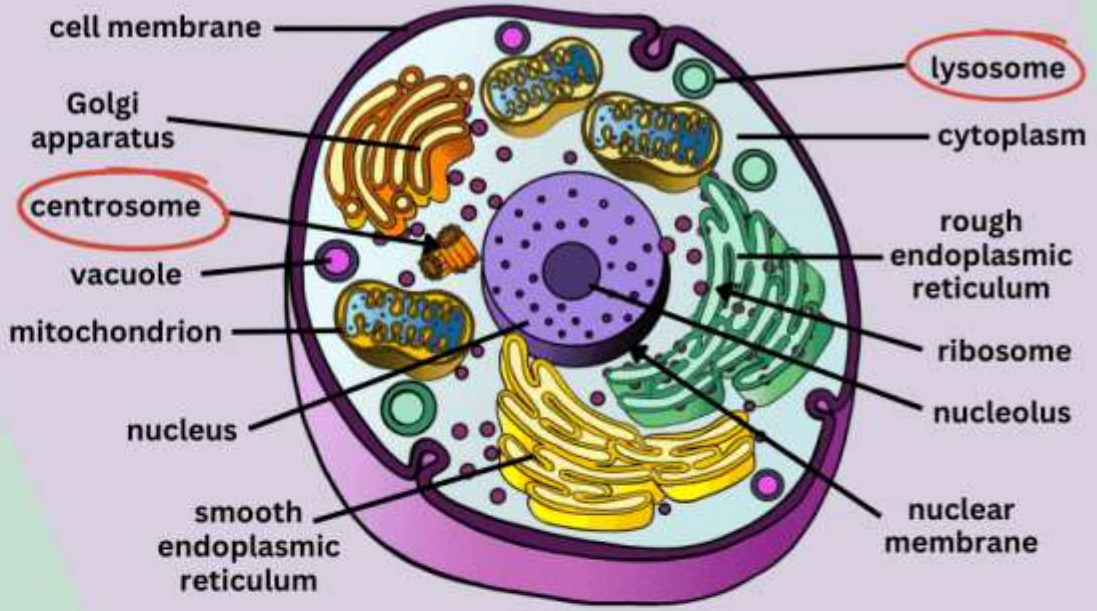
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PLANT CELL VS ANIMAL CELL

- Cell wall (rigid shape)
- Large central vacuole
- Chloroplasts (autotroph)
- Lack centrosomes
- Lack lysosomes

Plant Cell





Animal Cell

- No cell wall (irregular shape)
- Numerous small vacuoles
- No chloroplasts (heterotroph)
- Centrioles / centrosomes
- Lysosomes

sciencenotes.org

Functions of Cells

- Provide **structure and support**.
- Carry out **metabolism** (respiration, protein/lipid synthesis).
- **Growth and reproduction** (mitosis & meiosis).
- **Communication** through signaling molecules.
- Maintain **homeostasis** (balance inside the cell).

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TISSUES – *Group of Cells Working Together* – Animal Tissues

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Epithelial Tissue	Connective Tissue	Muscular Tissue	Nervous Tissue
• Covers body surfaces & lines organs. • Types: • Squamous	• Provides support, strength, and connection. • Types:	• Responsible for movement. • Types: • Skeletal (voluntary,	• Made of neurons. • Function: Receive and transmi

Plant Tissues

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Meristematic Tissue (growth tissues)

- Types:
- Types: Apical (tips of roots &
- Parenchyma – storage

Permanent Tissue

- **Collenchyma** – flexible support.
- **Sclerenchyma** – rigid support.

BIOMOLECULES

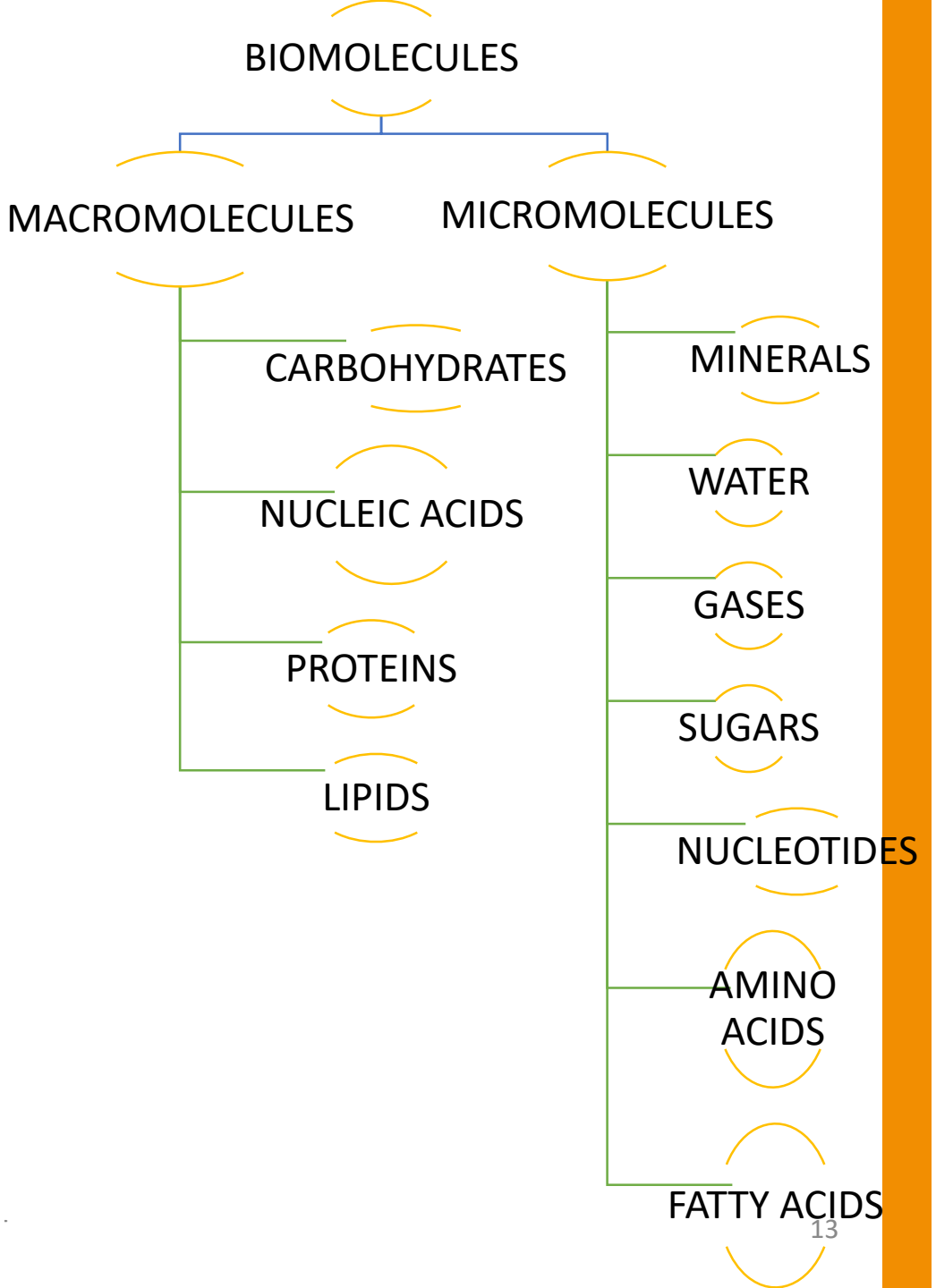
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- **What is a biomolecule?**
- Biomolecules, also known as biological molecules, are chemical compounds found in living organisms that are essential for the survival of living cells and play a vital role in all life processes.
- **Biomolecules** have a wide range of sizes and structures and perform a vast array of functions. The four major types of biomolecules are **carbohydrates, lipids, nucleic acids, and proteins**. Among biomolecules, nucleic acids, namely DNA and RNA, have the unique function of storing an organism's genetic code—the sequence of nucleotides that determines the amino acid sequence of proteins, which are of critical importance to life on Earth.
- **Biomolecules can be classified into two categories:**
- Macromolecules: These include protein, nucleic acid, and polysaccharides
- Micromolecules: These include minerals, water, gases, amino acids, nucleic acids, fatty acids and monosaccharides

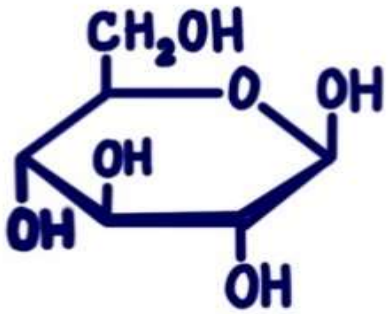
The major complex biomolecules of cells

Biomolecule	Building block (repeating unit)	Major functions
1. Protein	Amino acids	Fundamental basis of structure and function of cell (static and dynamic functions).
2. Deoxyribonucleic acid (DNA)	Deoxyribonucleotides	Repository of hereditary information.
3. Ribonucleic acid (RNA)	Ribonucleotides	Essentially required for protein biosynthesis.
4. Polysaccharide (glycogen)	Monosaccharides (glucose)	Storage form of energy to meet short term demands.
5. Lipid	Fatty acids, glycerol	Storage form of energy to meet long term demands; structural components of membranes.



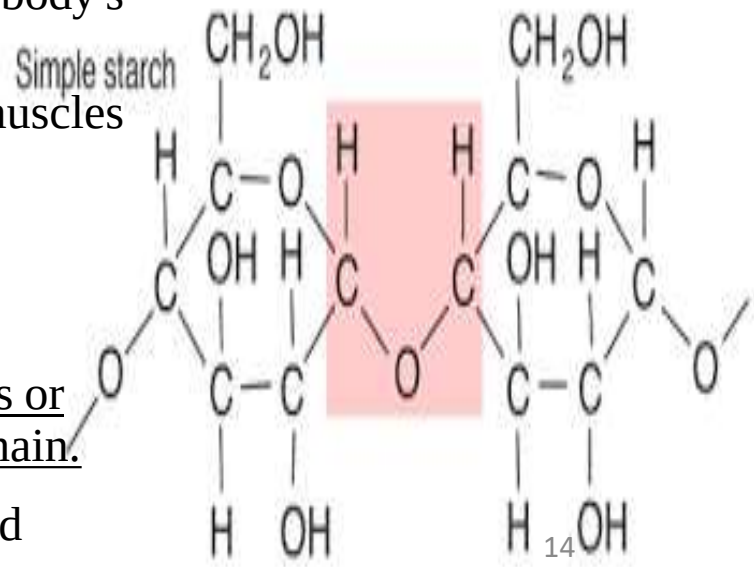
CARBOHYDRATES

- **Carbohydrates (hydrates of carbon) are a group of naturally occurring carbonyl compounds (aldehydes or ketones) that also contain several hydroxyl groups.** It may also include derivatives which produce such compounds on hydrolysis.
- They are the most abundant organic molecules in nature and are referred to as "saccharides".
- Simple carbohydrates which are soluble in water and sweet in taste called **"Sugars".**
  - They can be added to foods, such as the sugar in candy, dessert processed foods, and regular soda.
- Body breaks down carbohydrates into glucose.
- Glucose, or blood sugar, is the main source of energy for your body's cells, tissues, and organs.
- Glucose can be used immediately or stored in the liver and muscles for later use, as Glycogen.
- Carbohydrates consist of **carbon, hydrogen, and oxygen.**
- The general empirical structure for carbohydrates is $\mathbf{(CH_2O)_n}$.
- They are organic compounds organized in the form of aldehydes or ketones with multiple hydroxyl groups coming off the carbon chain.
- The building blocks of all carbohydrates are simple sugars called **monosaccharides**.



Glucose

$C_6H_{12}O_6$



FUNCTIONS OF CARBOHYDRATES

- Carbohydrates participate in a wide range of functions

1. They are the **most abundant dietary source of energy (4Cal/g)** for all organisms.
2. Carbohydrates are precursors for many organic compounds (fats, amino acids).
3. Carbohydrates (as glycoproteins and glycolipids) participate in the structure of cell membrane and cellular functions such as cell growth, adhesion and division.
4. They are structural components of many organisms. These include the fiber (cellulose) of plants, exoskeleton of some insects and the cell wall of microorganisms.
5. Carbohydrates also serve as the storage form of energy (glycogen) to meet the immediate energy demands of the body.
6. Ribose and deoxyribose sugars forms the structural frame of the genetic material, RNA and DNA.
7. Stored carbohydrates acts as energy source instead of proteins.
8. Carbohydrates are intermediates in biosynthesis of fats and proteins.
9. Carbohydrates aid in regulation of nerve tissue and is the energy source for brain.
10. Carbohydrates gets associated with lipids and proteins to form surface antigens, receptor molecules, vitamins and antibiotics.
11. They form structural and protective components, like in cell wall of plants and microorganisms.
12. In animals they are important constituent of connective tissues.
13. They participate in biological transport, cell-cell communication and activation of growth factors.
14. Carbohydrates that are rich in fibre content help to prevent constipation.
15. Also they help in modulation of immune system.

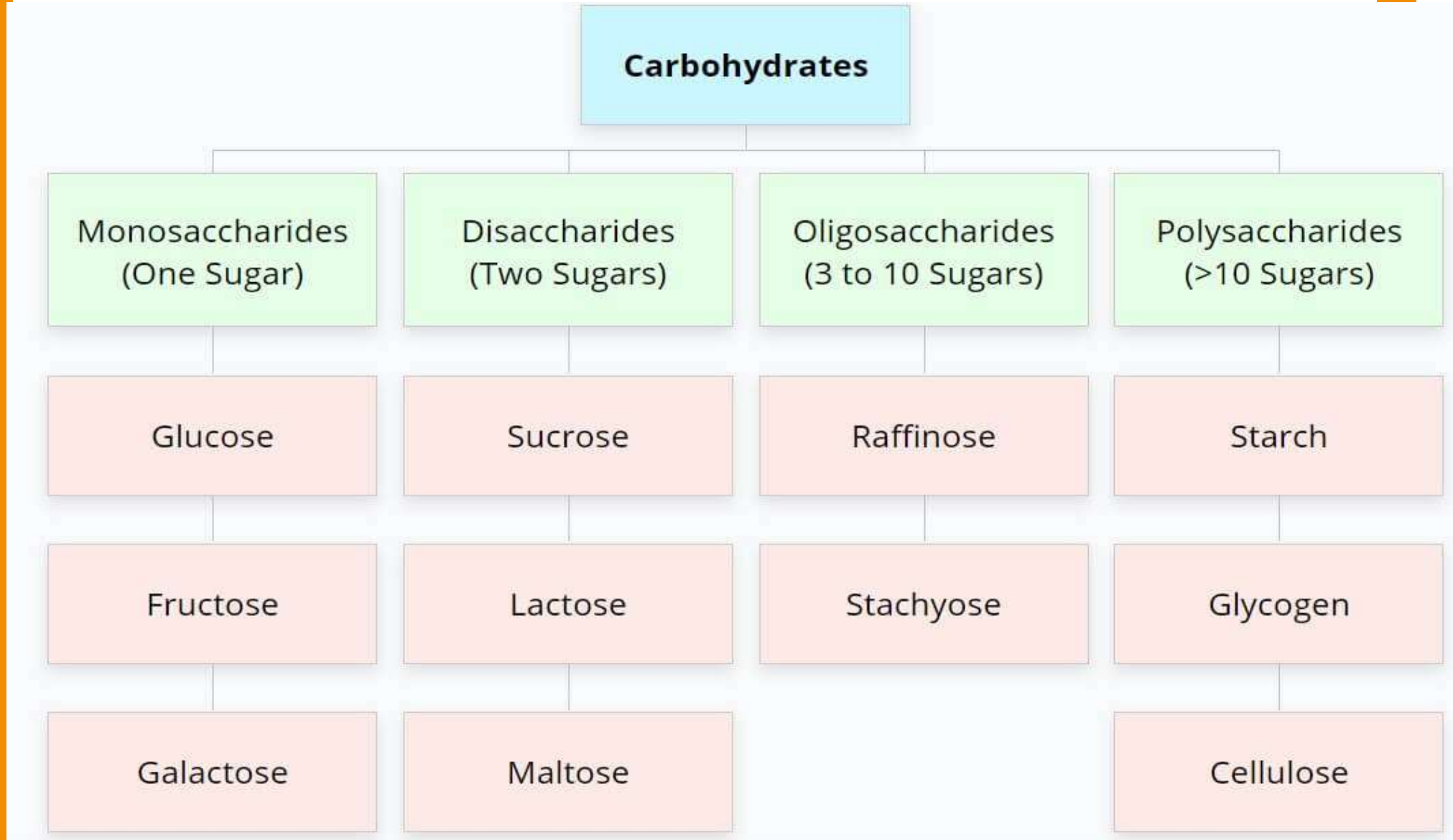
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CLASSIFICATION OF CARBOHYDRATES

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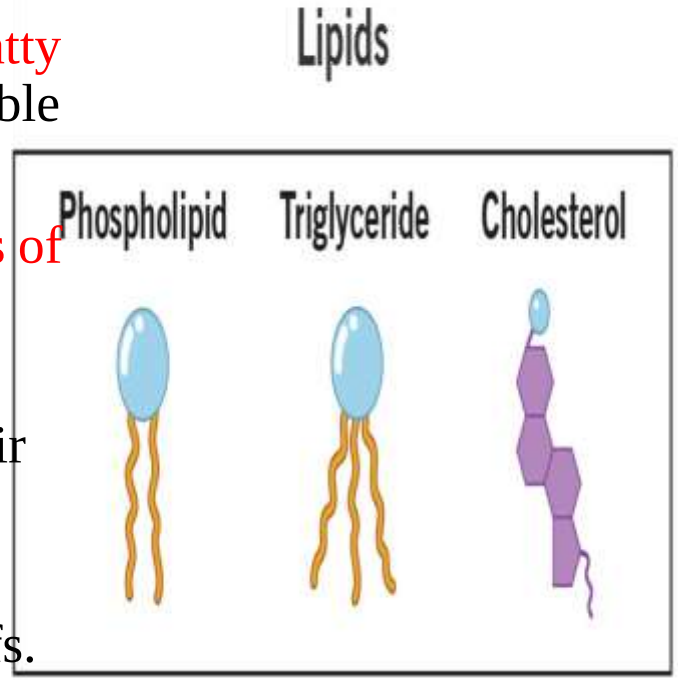
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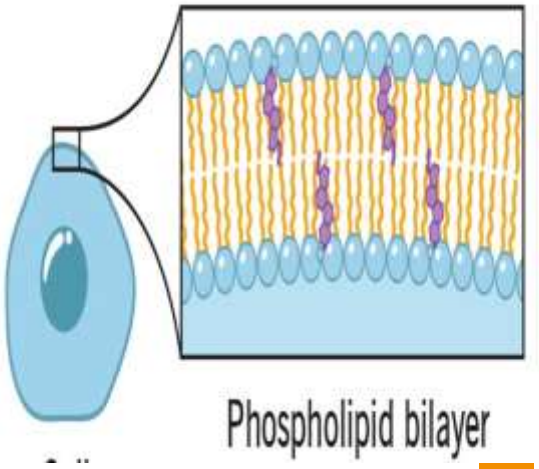
CLASSIFICATION OF CARBOHYDRATES

- **Monosaccharides**
  - Monosaccharides, or simple sugars, consist of a single polyhydroxy aldehyde or ketone unit. The most abundant monosaccharide in nature is the six-carbon sugar D-Glucose.
  - Example: Glucose, Fructose, Galactose
- **Oligosaccharides**
  - Oligosaccharides consist of short chains (2-10) of monosaccharide units, or residues, joined by characteristic linkages called glycosidic bonds. The most abundant are the disaccharides, with two monosaccharide units.
  - Example: sucrose (cane sugar), lactose, maltose
- **Polysaccharides**
  - The polysaccharides are sugar polymers containing more than 10 or so monosaccharide units, and some have hundreds or thousands of units.
  - Example: starch, glycogen, cellulose, chitin, hyaluronic acid, heparin
  - Polysaccharides are of two types based on their function and composition. Based on function, polysaccharides of two types storage and structural.
    - A. Storage polysaccharide – starch, glycogen
    - B. Structural polysaccharide – cellulose, chitin
  - They are further classified depending on the type of molecules produced as a result of hydrolysis.
  - They may be homopolysaccharides, containing monosaccharides of the same type or heteropolysaccharides i.e., monosaccharides of different types.
    - Examples of Homopolysaccharides are starch, glycogen, cellulose, pectin.
    - Heteropolysaccharides are Hyaluronic acid, Chondroitin.

LIPIDS

- Lipids are a group of diverse macromolecules consisting of fatty acids and their derivatives that are insoluble in water but soluble in organic solvents.
- Lipids consist of fats, oils, hormones, and certain components of membranes that are grouped together because of their hydrophobic interactions.
- The lipids are essential constituents of the diet because of their high energy value.
- These are also essential for the fat-soluble vitamins and the essential fatty acids found with the fat of the natural foodstuffs.
- Fats combined with proteins (lipoproteins) are essential constituents of the cell membranes and mitochondria of the cell.
- Lipids occur naturally in living beings like plants, animals, and microorganisms that form various components like cell membranes, hormones, and energy storage molecules.
- Lipids exist in either liquid or non-crystalline solids at room temperatures and are colorless, odorless, and tasteless.
- These are composed of fatty acids and glycerol.



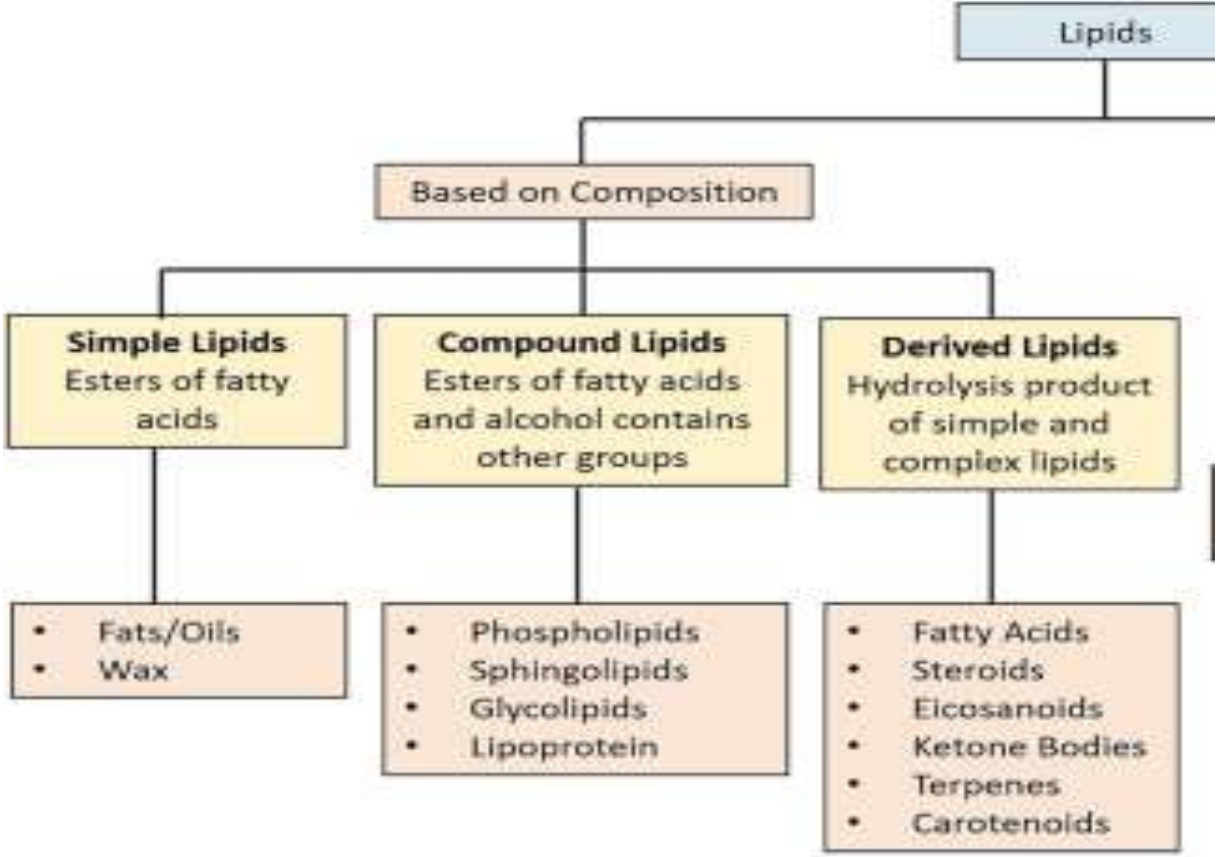


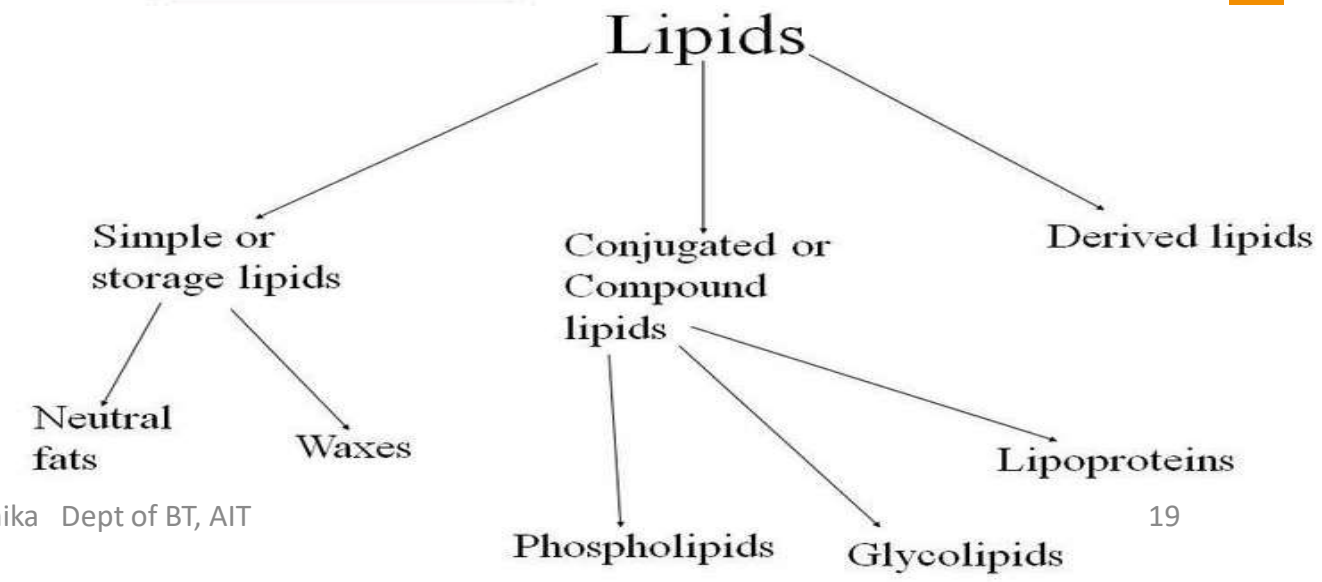
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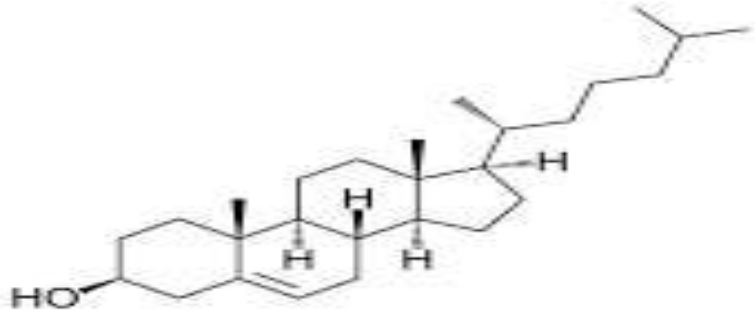
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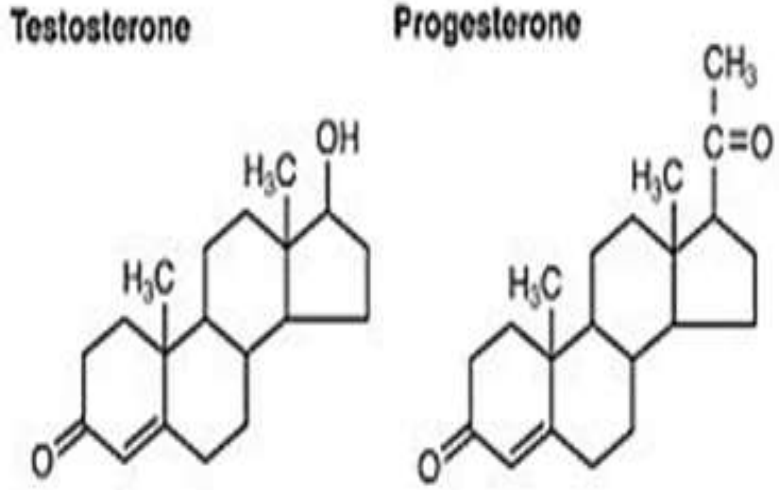
Derived lipids:

- Hydrolysis product of simple and compound lipids is called derived lipids.
- They include fatty acids, glycerol, sphingosine, terpenoid and steroid derivatives.
- Steroid derivatives are phenanthrene structures that are quite different from lipids made up of fatty acids.
- Steroid hormones are a group of hormones derived from cholesterol that act as chemical messengers in the body. The steroid hormones regulate many physiologic processes, including the development and function of the reproductive system.
- Examples-
  - Steroids – Cholesterol, ergosterol
  - Terpenes – Carotenoids

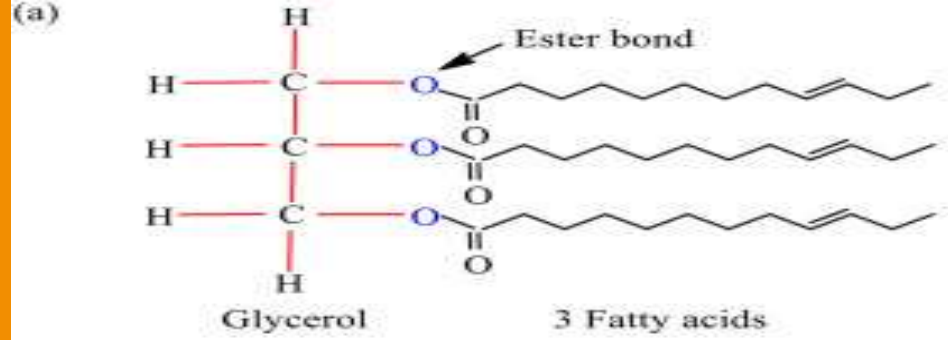


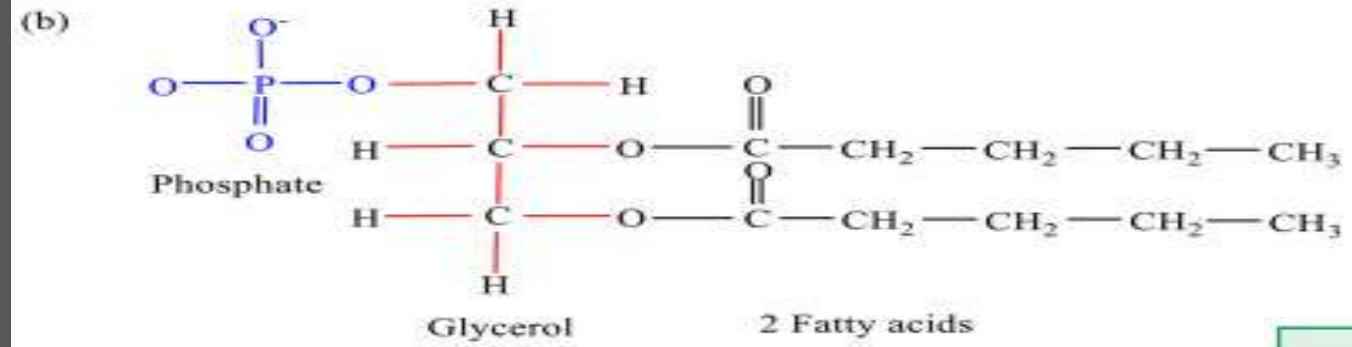
Cholesterol

$C_{27}H_{46}O$



Steroid Hormones





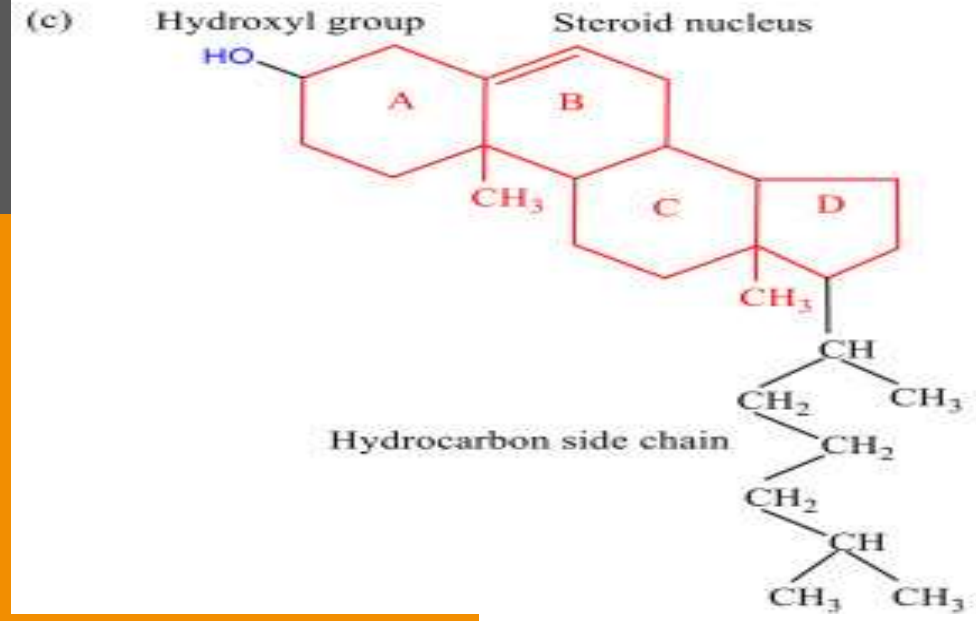
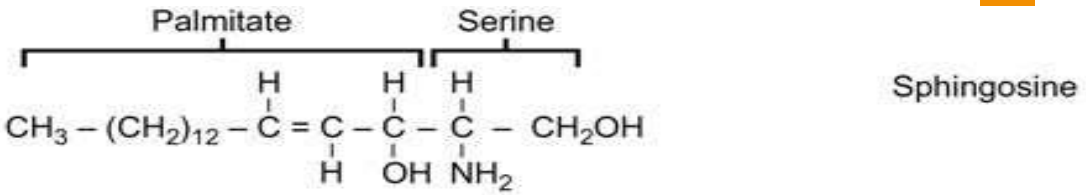


Fig.1 (a) Triglyceride, (b) Phospholipid, (c) Cholesterol



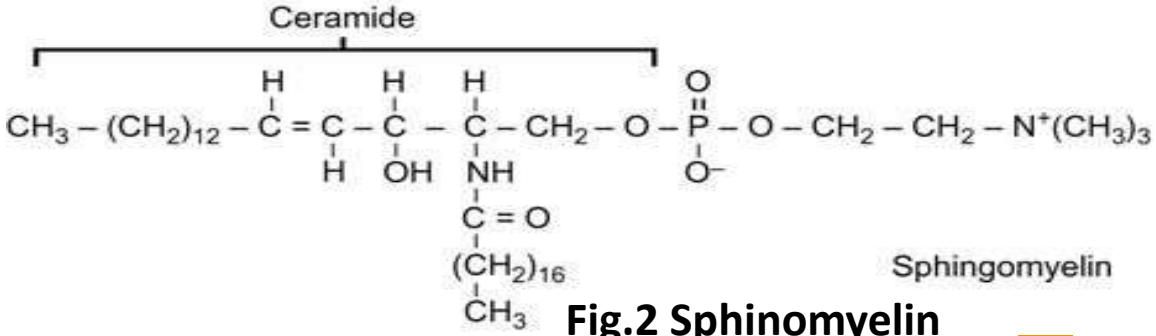


Fig.2 Sphinomyelin

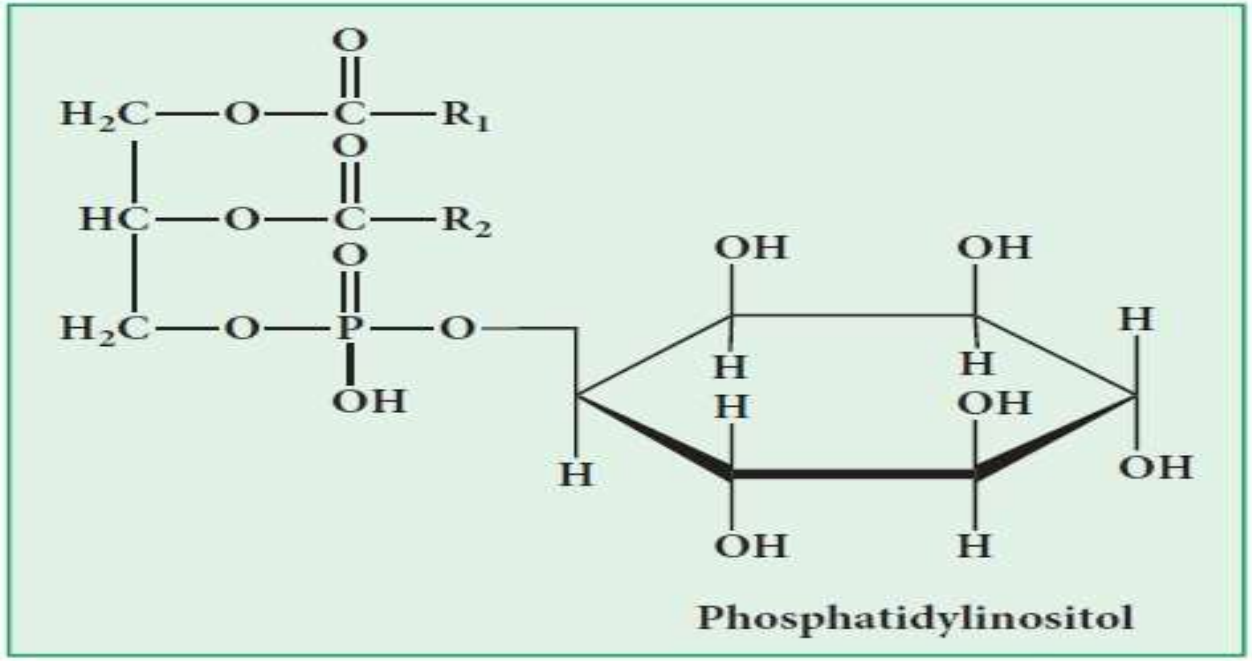


Figure 2.11 Structure of phosphatidylinositol

PROPERTIES OF LIPIDS

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- Lipids may be either liquids or non-crystalline solids at room temperature.
- Pure fats and oils are colorless, odorless, and tasteless.
- They are energy-rich organic molecules
- Insoluble in water
- Soluble in organic solvents like alcohol, chloroform, acetone, benzene, etc.
- No ionic charges
- Solid triglycerols (Fats) have high proportions of saturated fatty acids.
- Liquid triglycerols (Oils) have high proportions of unsaturated fatty acids.

FUNCTIONS OF LIPIDS

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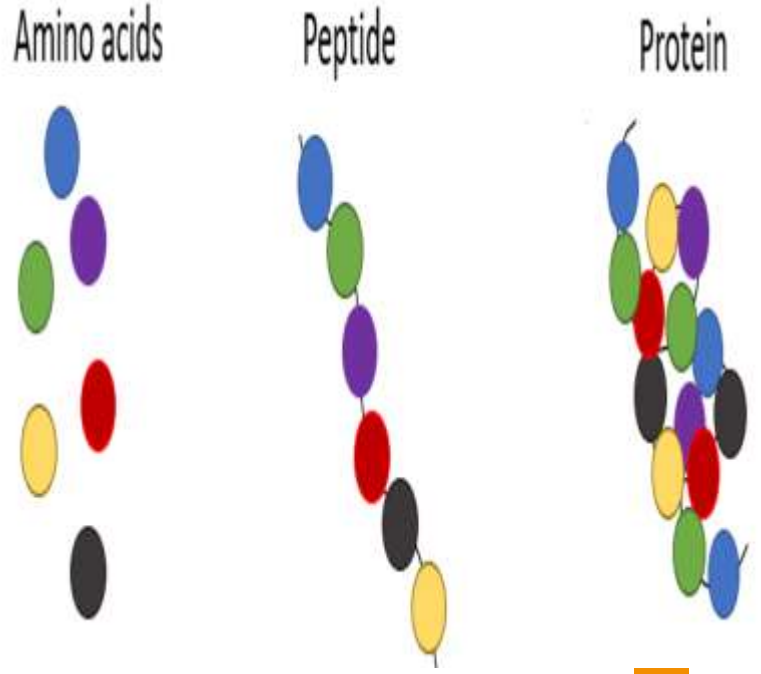
- Biological lipids are a chemically diverse group of compounds, and the biological functions of the lipids are as diverse as their chemistry.
- In the body, fats serve as an efficient source of energy and are also stored in the adipose tissues.
- These also serve as an insulating material in the subcutaneous tissues and around certain organs give shape and smooth appearance to the body.
- Phospholipids and sterols are major structural elements of biological membranes.
- Lipids also act as the structural component of the cell and provide the hydrophobic barrier that allows the separation of the aqueous contents of the cell and subcellular structures.
- Other lipids, although present in relatively small quantities, play crucial roles as enzyme cofactors, electron carriers, light-absorbing pigments, and hydrophobic anchors for proteins.
- Lipids are important as cellular metabolic regulators (steroid hormones and prostaglandins).
- They serve as a source of fat soluble vitamins (A, D, E and K) and hormones.
- Lipids are also activators of enzymes like glucose-6-phosphatase, β-hydroxybutyric dehydrogenase, and stearyl CoA desaturase.

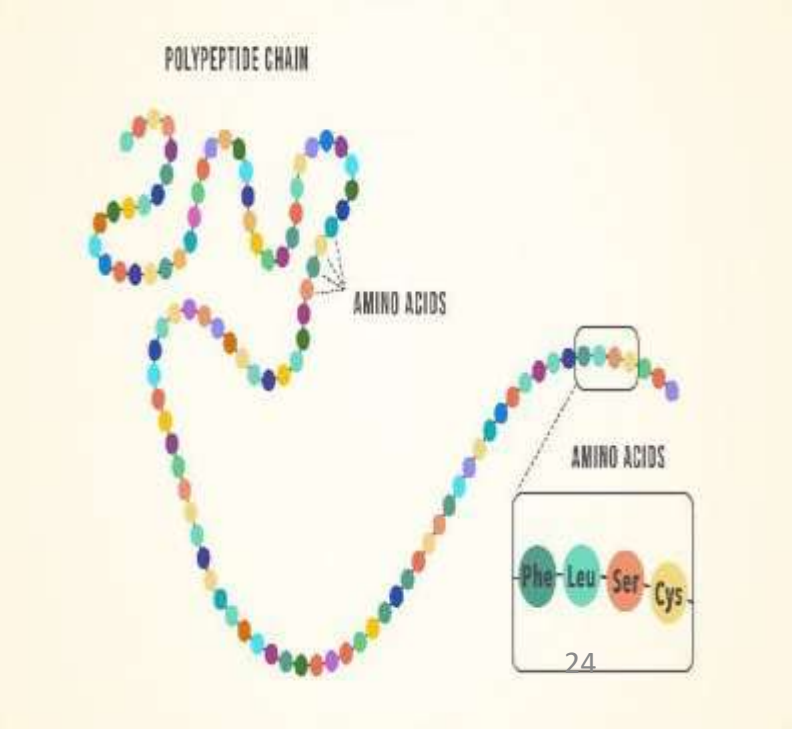
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PROTEINS

- Proteins are nitrogen containing, most abundant organic macromolecules widely distributed in animals and plants. They perform structural and dynamic functions in the organisms.
- Proteins are species-specific and are unique to each organism. Similarly, these are also organ-specific in that the proteins of the brain are different from those in the muscles.
- Proteins are also termed polypeptides, as they are long chains of amino acids connected by peptide bonds.
- Proteins are polymers composed of L-α-amino acids. They are 20 in number and classified into different groups based on their structure, chemical nature, nutritional requirement and metabolic fate.
- Proteins can be classified in three ways based on their --
  - a) Function
  - b) Chemical nature and Solubility
  - c) Nutritional value



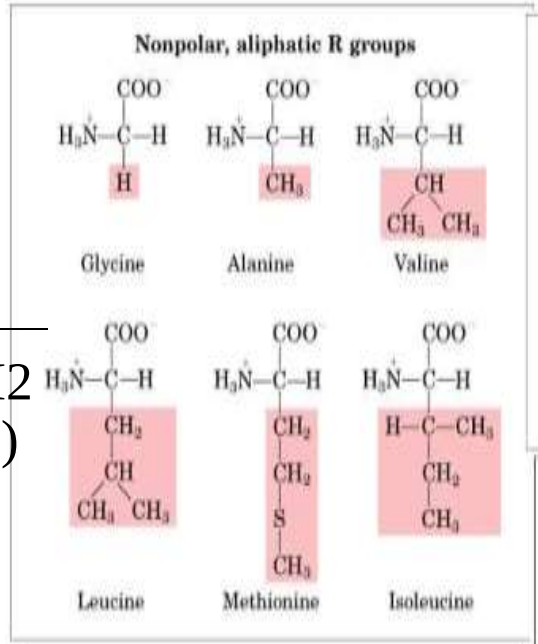


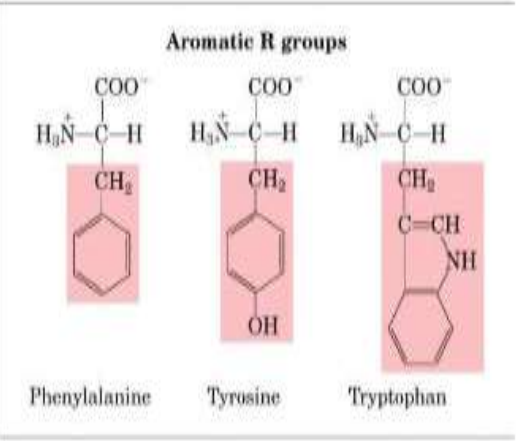
AMINO ACIDS

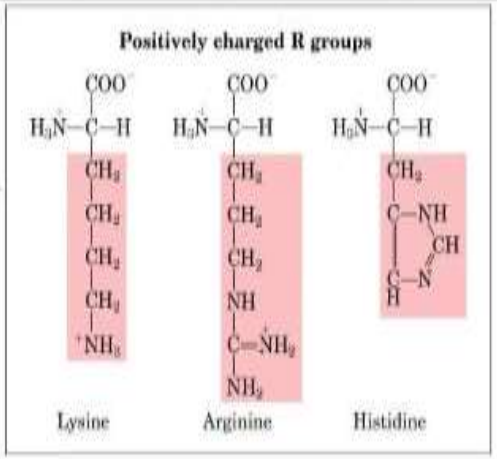
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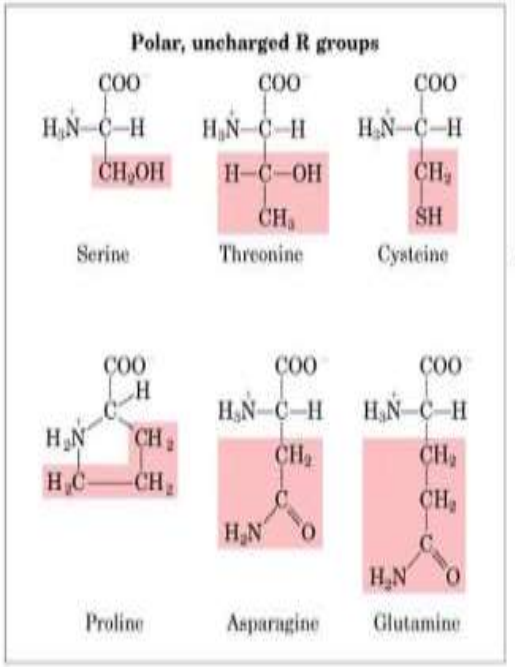
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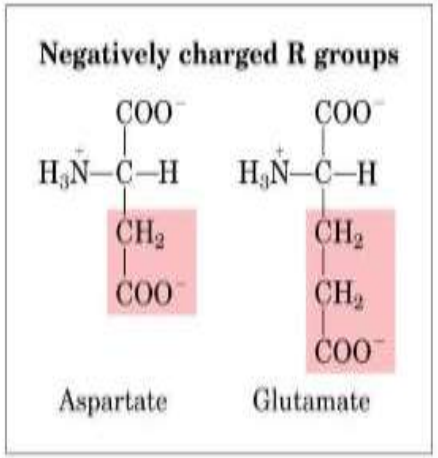
- **Amino acids** are the monomers that make up proteins. They are a group of organic compounds containing two functional groups—amino and carboxyl. The amino group (—NH2) is basic while the carboxyl group (—COOH) is acidic in nature.
- **General structure of amino acids**
- The amino acids are termed as α-amino acids, if both the carboxyl and amino groups are attached to the same carbon atom.
- The α-carbon atom binds to a side chain represented by R (another atom or group of atoms bonded to the central atom) which is different for each of the 20 amino acids found in proteins.
- The amino acids mostly exist in the ionized form in the biological system.











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FUNCTIONS OF PROTEINS

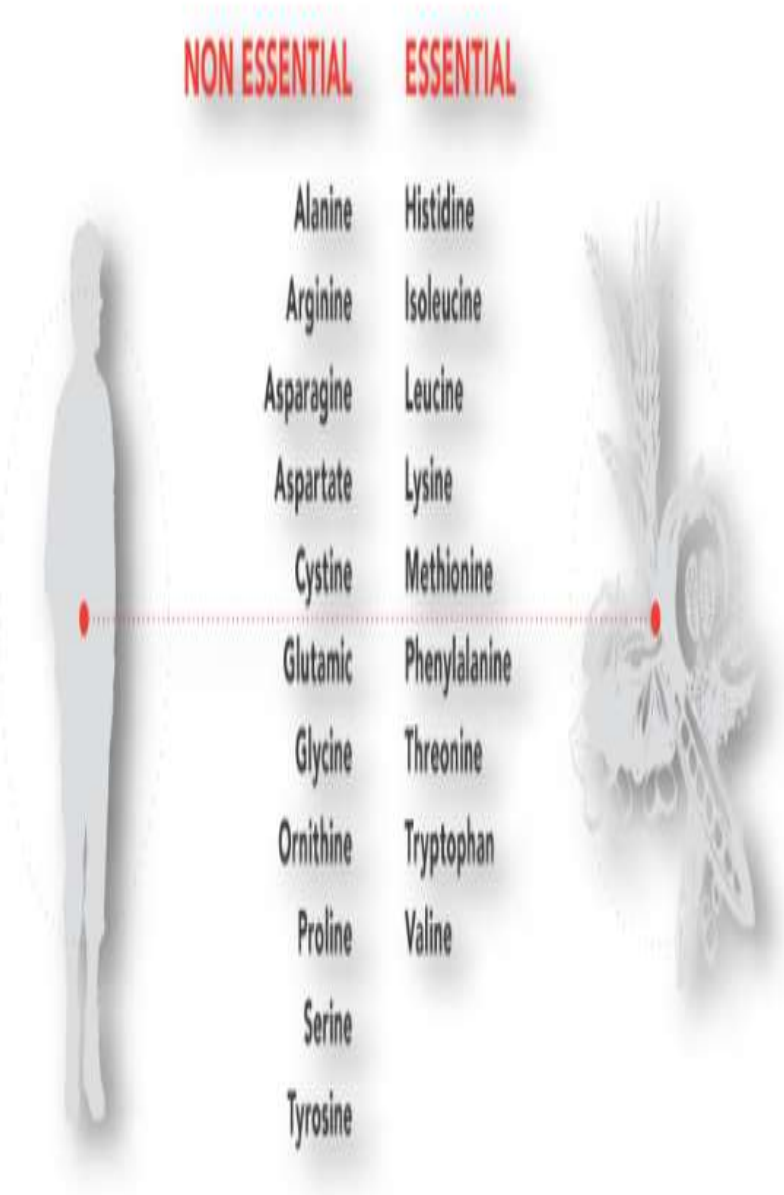
- Proteins perform a great variety of specialized and essential functions in the living cells.
- These functions may be broadly grouped as static (structural) and dynamic.
- Structural functions
  - Certain proteins perform brick and mortar roles and are primarily responsible for structure and strength of body.
  - These include collagen and elastin found in bone matrix, vascular system and other organs and α-keratin present in epidermal tissues.
- Dynamic functions
  - The dynamic functions of proteins are more diversified in nature.
  - These include proteins acting as enzymes, hormones, blood clotting factors, immunoglobulins, membrane receptors, storage proteins, besides their function in genetic control, muscle contraction, respiration etc.
- Proteins performing dynamic functions are appropriately regarded as the working horses of cell.

NUTRITIONAL CLASSIFICATION OF PROTEINS

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- The nutritive value of proteins is determined by the composition of essential amino acids.
- From the nutritional point of view, proteins are classified into 3 categories.
- 1. Complete proteins : These proteins have all the ten essential amino acids in the required proportion by the human body to promote good growth.
  - e.g. egg albumin, milk casein.
- 2. Partially incomplete proteins : These proteins partially lack one or more essential amino acids, and can promote moderate growth.
  - e.g. wheat and rice proteins (limiting Lys, Thr).
- 3. Incomplete proteins : These proteins completely lack or or more essential amino acids. Hence they do not promote growth at all.
  - e.g. gelatin (lacks Trp), zein (lacks Trp, Lys).



FUNCTIONS OF PROTEINS

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1. Proteins are essential biomolecules that are critical to life and to perform various activities. Some of the important biological roles of proteins are:
2. Many proteins act as catalysts that enhance the rate of chemical reactions in various metabolic pathways. -- ENZYMES
3. The fibrous proteins are a component of various tissues holding the skeletal elements together like collagen, which is a structural unit of connective tissues.
4. The nucleoproteins serve as carriers of genetic characters and hence govern the inheritance of traits.
5. Proteins also perform transport functions that regulate the transport of many compounds in and out of the cells and accumulate inside at much higher concentrations than expected from diffusion alone.
6. Various protein hormones regulate the growth of plants and animals, besides controlling many other physiological functions.
7. Blood plasma has multiple soluble proteins that can be used for the treatment of shock produced by severe injuries and operations.
8. Interferons are regulatory glycoproteins produced by many eukaryotic cells in response to virus infection, endotoxins, antigenic stimuli, and many parasitic organisms.
9. Peptides from humans called defensins are antibiotic in nature.

Brief introduction of Structure and functions of the human organ system

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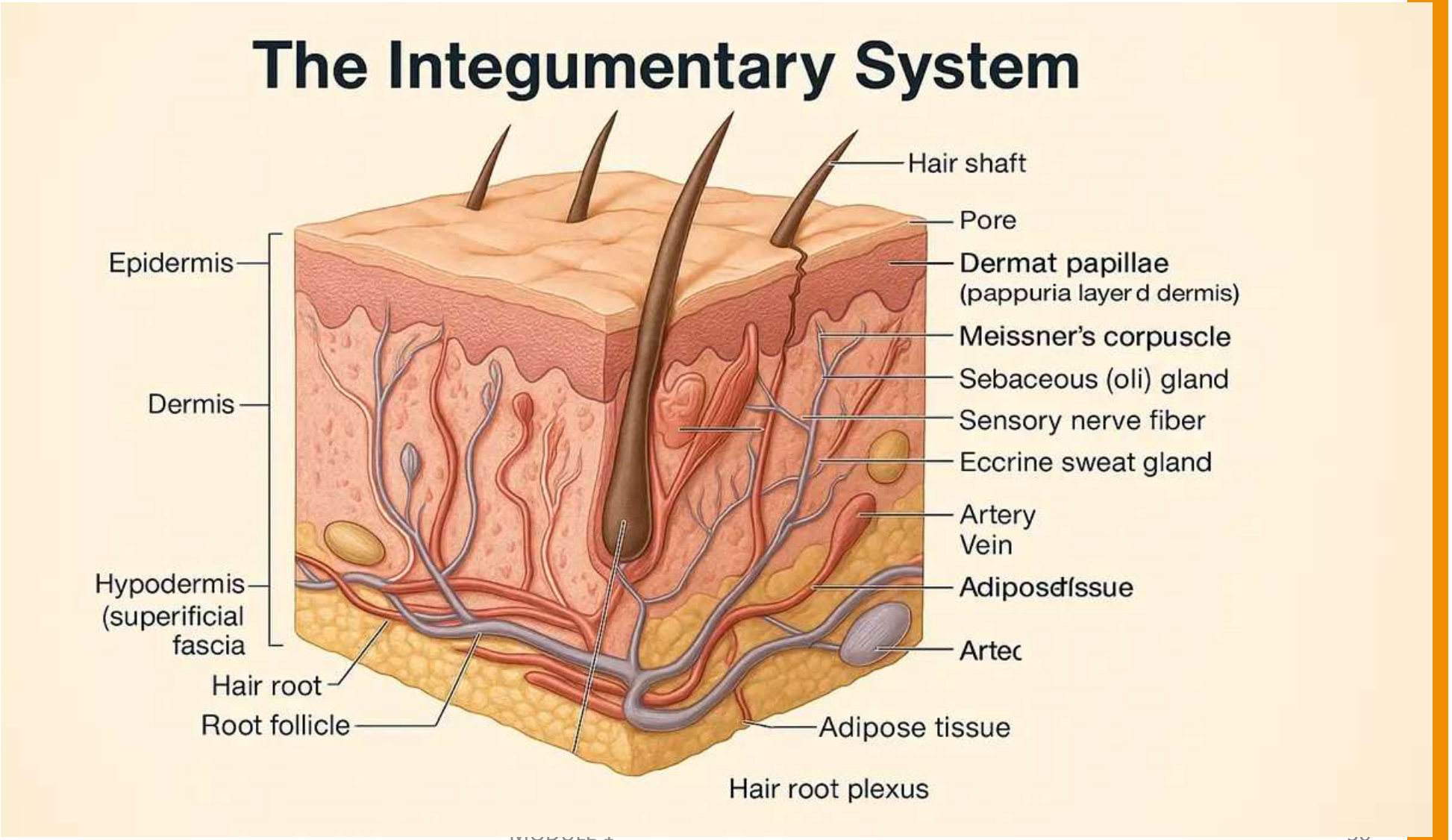
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1. Integumentary system - skin and its appendages; hair, nails, sweat glands and oil glands.
2. Skeletal system – Bones, Cartilage, etc
3. Muscular system – Skeletal, Smooth and Cardiac Muscles
4. Nervous System – Brain, Spinal cord, nerves
5. Circulatory/Cardiovascular system – Heart, Arteries, Veins, Capillaries
6. Lymphatic system – Lymph nodes, lymph vessels, and organs (such as the spleen and thymus)
7. Respiratory system – Lungs, trachea, bronchi, and diaphragm.
8. Digestive system – Mouth, esophagus, stomach, intestines
9. Excretory system – Kidneys, nephrons, ureters, bladder, and urethra
10. Endocrine System – Glands such as pituitary, thyroid, adrenal, and reproductive glands
11. Reproductive System – Males and females – ovaries, uterus, fallopian tubes, vagina, testes, seminal vesicles, prostate, and penis.

Integumentary System

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Structure of the Integumentary System

- 1. Skin (Primary organ)
- The skin has three main layers:
  - Epidermis (outer layer)
    - Made of stratified squamous epithelial cells.
    - Contains keratinocytes (produce keratin for waterproofing), melanocytes (produce melanin for skin color and UV protection), and Langerhans cells (immune defense).
    - Constantly regenerates.
  - Dermis (middle layer)
    - Made of connective tissue.
    - Contains blood vessels, nerves, sweat glands, sebaceous glands, and hair follicles.
    - Provides strength, elasticity, and sensation.
  - Hypodermis (subcutaneous layer)
    - Made of loose connective tissue and fat.
    - Provides insulation, shock absorption, and energy storage.
- 2. Accessory Structures
- Hair – Provides protection (scalp, eyelashes, eyebrows), reduces heat loss, and enhances touch sensation.
- Nails – Protect tips of fingers and toes, aid in grasping objects.
- Glands – Sweat glands (sudoriferous glands): Help regulate temperature and excrete waste (salts, water, urea).
- Sebaceous glands: Secrete sebum (oil) to lubricate skin and hair.
- Ceruminous glands: Produce earwax for ear protection.

Functions of the Integumentary System

1. Protection – Shields body from mechanical damage, pathogens, UV radiation, and chemical exposure.
2. Regulation of Body Temperature – Sweat and blood flow help maintain temperature (thermoregulation).
3. Sensation – Nerve endings in skin detect touch, pain, temperature, and pressure.
4. Excretion – Sweat glands eliminate salts, water, and waste products.
5. Vitamin D Synthesis – Skin produces vitamin D in presence of sunlight, important for bone health.
6. Waterproofing – Keratin prevents excessive water loss or entry.
7. Immunity – Acts as a first line of defense and houses immune cells.

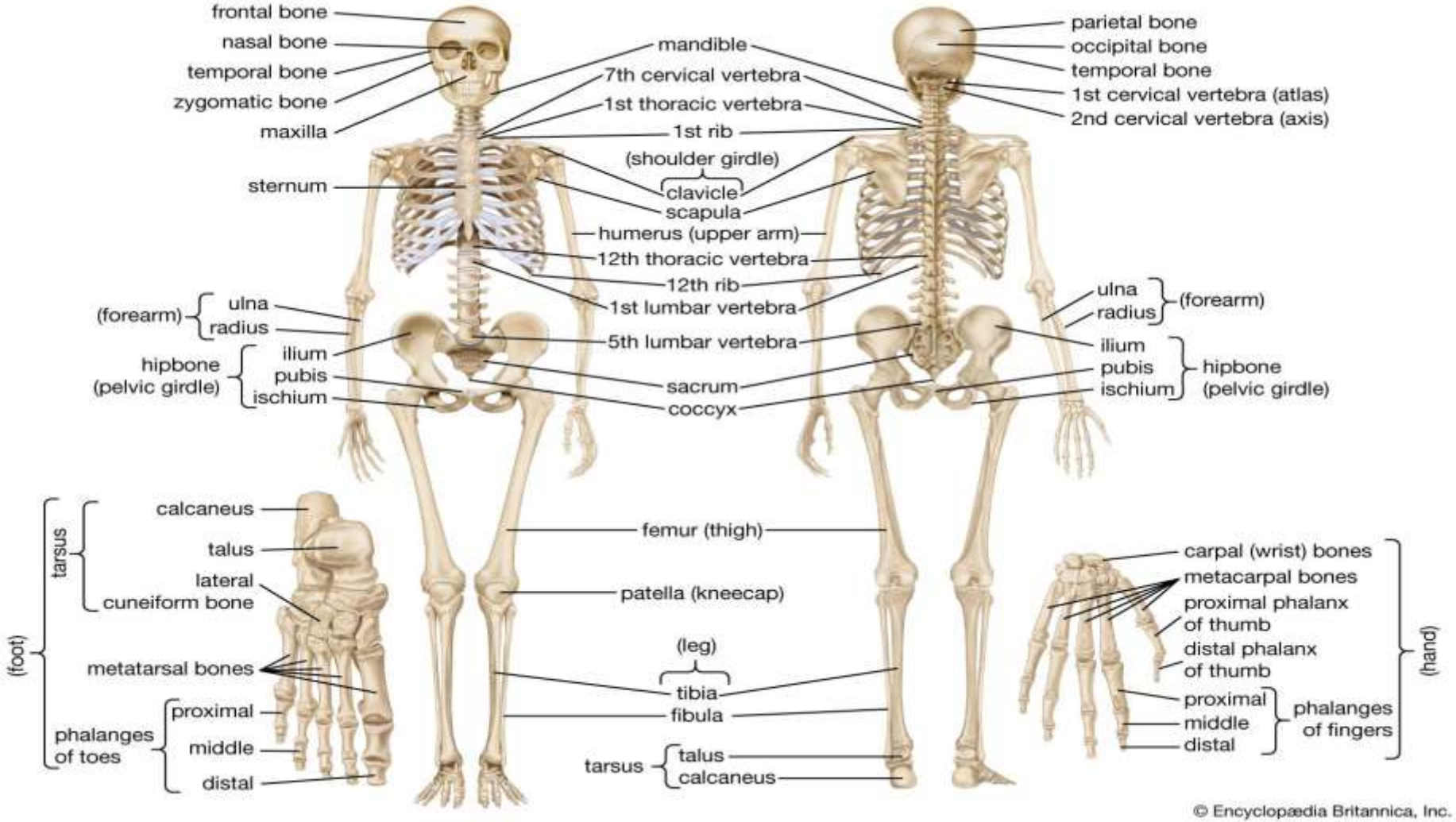
Clinical Importance / Disorders

- Acne: Caused by blocked sebaceous glands.
- Psoriasis: Autoimmune condition causing rapid skin cell turnover.
- Eczema: Inflammatory skin condition.
- Skin Cancer: Melanoma, basal cell carcinoma, squamous cell carcinoma.
- Infections: Fungal (ringworm), bacterial (impetigo), viral (warts, herpes).

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THE SKELETAL SYSTEM



Structure of the Skeletal System

- **1. Bones**
- Hard structures made of **calcium, phosphate, and collagen**.
- Types of bones:
  - **Long bones** – e.g., femur, humerus (support weight and movement).
  - **Short bones** – e.g., carpals, tarsals (stability, little movement).
  - **Flat bones** – e.g., skull, ribs (protection, surface for muscles).
  - **Irregular bones** – e.g., vertebrae (complex shapes).
  - **Sesamoid bones** – e.g., patella (embedded in tendons).
- **2. Cartilage**
- Smooth, flexible connective tissue.
- Found in joints, nose, ear, trachea.
- Prevents friction between bones.

Structure of the Skeletal System

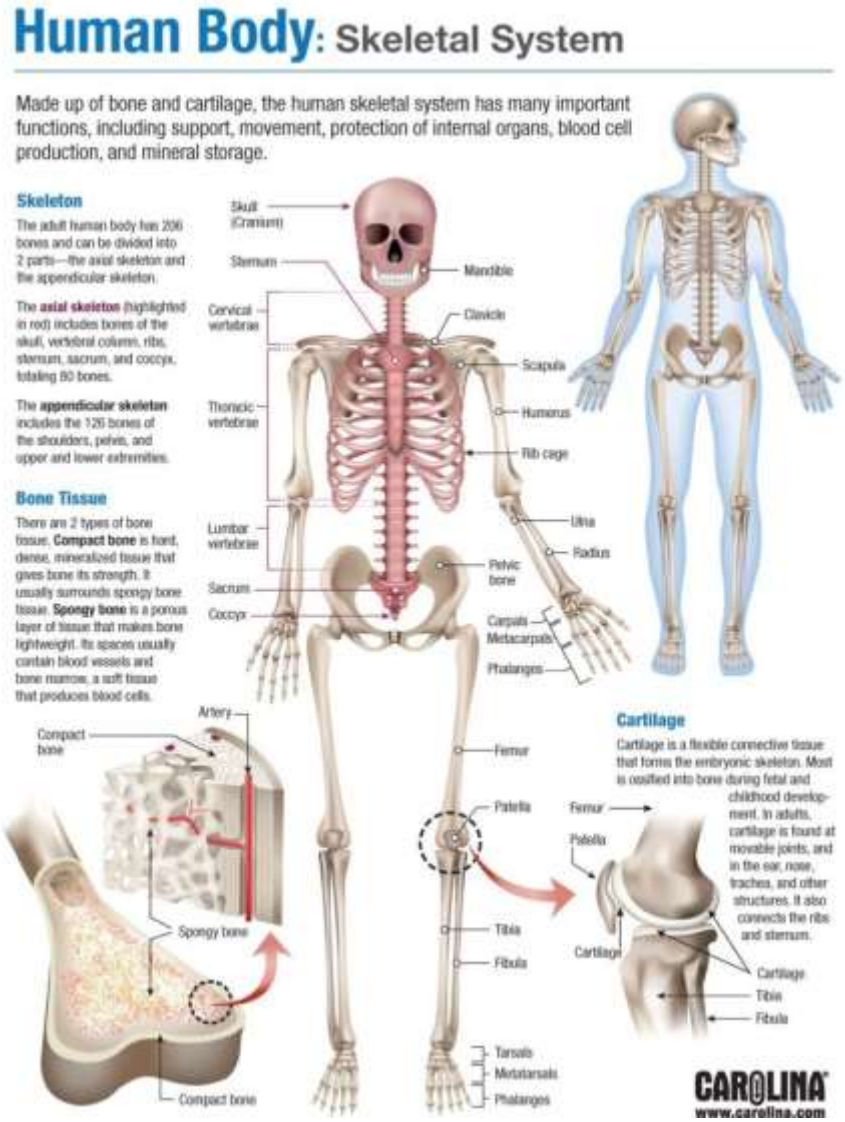
- **3. Ligaments**
- Strong bands of connective tissue.
- Connect **bone to bone**.
- Provide stability to joints.
- **4. Joints**
- Points where two or more bones meet.
- Types of joints:
  - **Fixed joints** (immovable, e.g., skull sutures).
  - **Slightly movable joints** (e.g., vertebrae).
  - **Freely movable joints (synovial):**
    - Hinge (elbow, knee)
    - Ball-and-socket (shoulder, hip)
    - Pivot (neck – atlas and axis)
    - Gliding (wrist, ankle)
    - Saddle (thumb)

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Divisions of the Human Skeleton

- **Axial Skeleton** (80 bones)
  - Skull, vertebral column, rib cage.
  - Supports and protects vital organs (brain, heart, lungs).
- **Appendicular Skeleton** (126 bones)
  - Limbs and girdles (shoulder & pelvic).
  - Enables movement and locomotion.



Functions of the Skeletal System

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- **Support:** Provides framework and shape for the body.
- **Protection:** Shields delicate organs (e.g., skull protects brain, rib cage protects heart and lungs).
- **Movement:** Works with muscles for motion (musculoskeletal system).
- **Mineral Storage:** Stores calcium and phosphorus.
- **Blood Cell Formation (Hematopoiesis):** Occurs in red bone marrow.
- **Fat Storage:** Yellow marrow stores fat as an energy reserve.

Disorders of the Skeletal System

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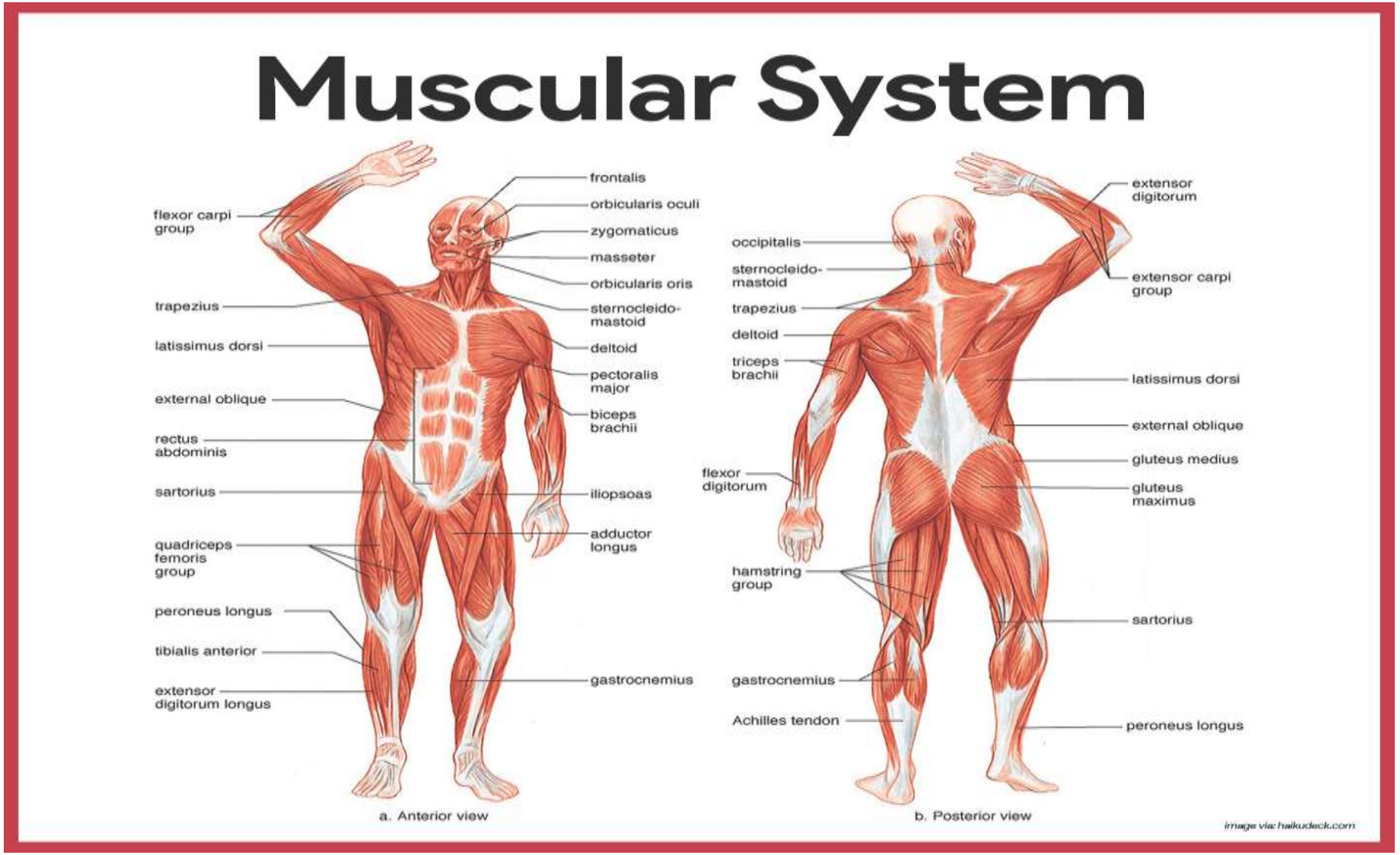
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- **Fractures:** Breaks in bones due to injury.
- **Osteoporosis:** Weakening of bones due to loss of calcium (common in elderly, especially women).
- **Arthritis:** Inflammation of joints.
- **Rickets:** Softening of bones due to Vitamin D deficiency (in children).
- **Scoliosis/Kyphosis/Lordosis:** Abnormal curvatures of the spine.
- **Osteomyelitis:** Bone infection caused by bacteria.

THE MUSCULAR SYSTEM

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Types of Muscles

- **Skeletal Muscles**
  - **Structure:** Long, cylindrical, multinucleated, striated (striped).
  - **Control:** Voluntary (under conscious control).
  - **Functions:** Body movement, posture maintenance, facial expressions, locomotion.
  - **Examples:** Biceps, triceps, quadriceps, diaphragm.
- **Smooth Muscles**
  - **Structure:** Spindle-shaped, single nucleus, non-striated.
  - **Control:** Involuntary (not under conscious control).
  - **Functions:** Found in walls of internal organs (stomach, intestines, blood vessels). Helps in peristalsis, regulation of blood flow, and movement of food through digestive tract.
- **Cardiac Muscle**
  - **Structure:** Branched, striated, single nucleus per cell, interconnected by **intercalated discs**.
  - **Control:** Involuntary.
  - **Functions:** Found only in the heart; pumps blood continuously throughout life.

Functions of the Muscular System

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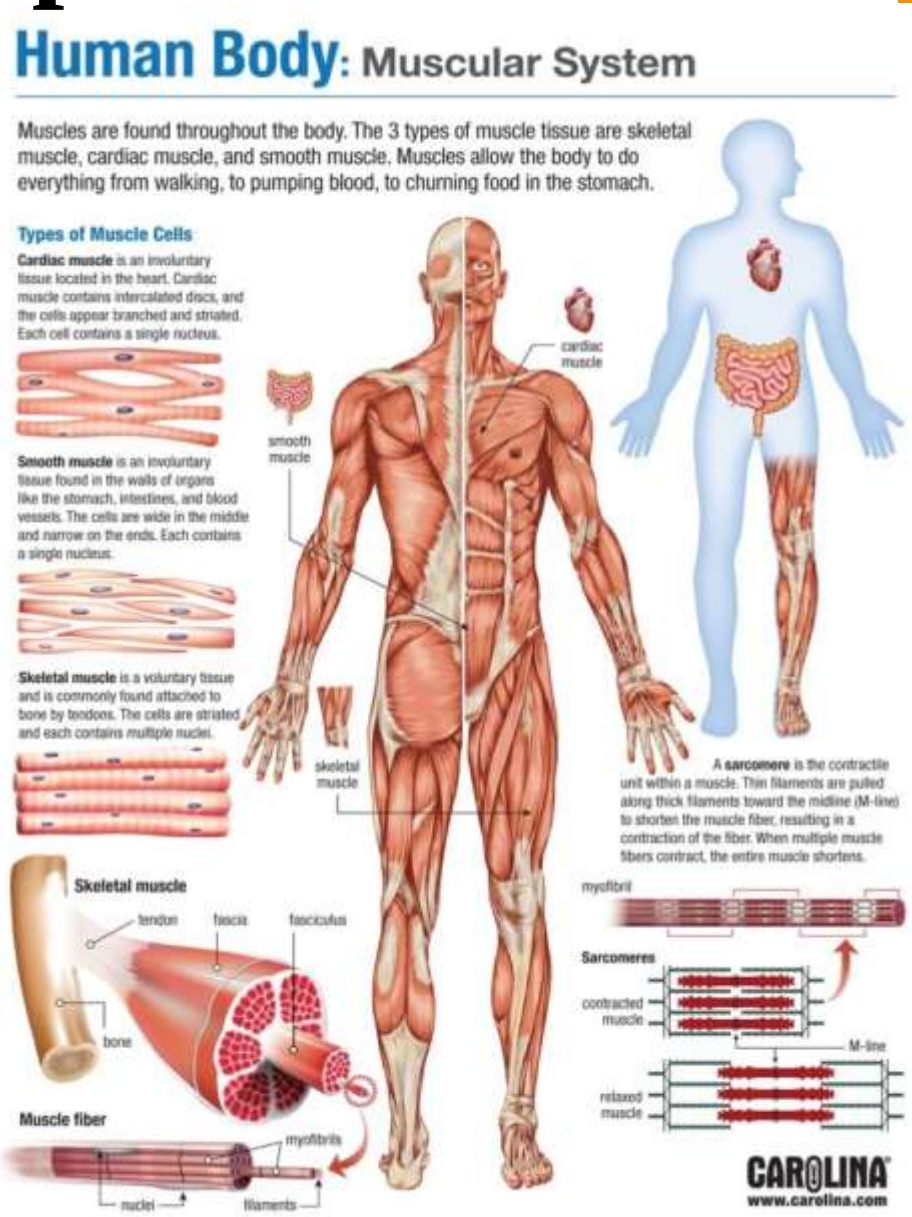
- **Movement:** Works with skeleton for locomotion and manipulation of environment.
- **Posture Maintenance:** Keeps the body upright and stable.
- **Heat Production:** Muscle contractions generate heat (important for thermoregulation).
- **Circulation:** Cardiac muscle pumps blood; skeletal muscles assist in venous return.
- **Digestion & Other Functions:** Smooth muscles move food, regulate airways, control pupil size, etc.

Major Muscle Groups

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- **Head & Neck:** Masseter (chewing), Sternocleidomastoid (neck movement).
- **Upper Body:** Pectoralis major (chest), Biceps brachii (arm), Trapezius (shoulders/back).
- **Lower Body:** Quadriceps (front thigh), Hamstrings (back thigh), Gastrocnemius (calf), Gluteus maximus (hip).
- **Core:** Rectus abdominis (abs), Obliques (sides).



Disorders of the Muscular System

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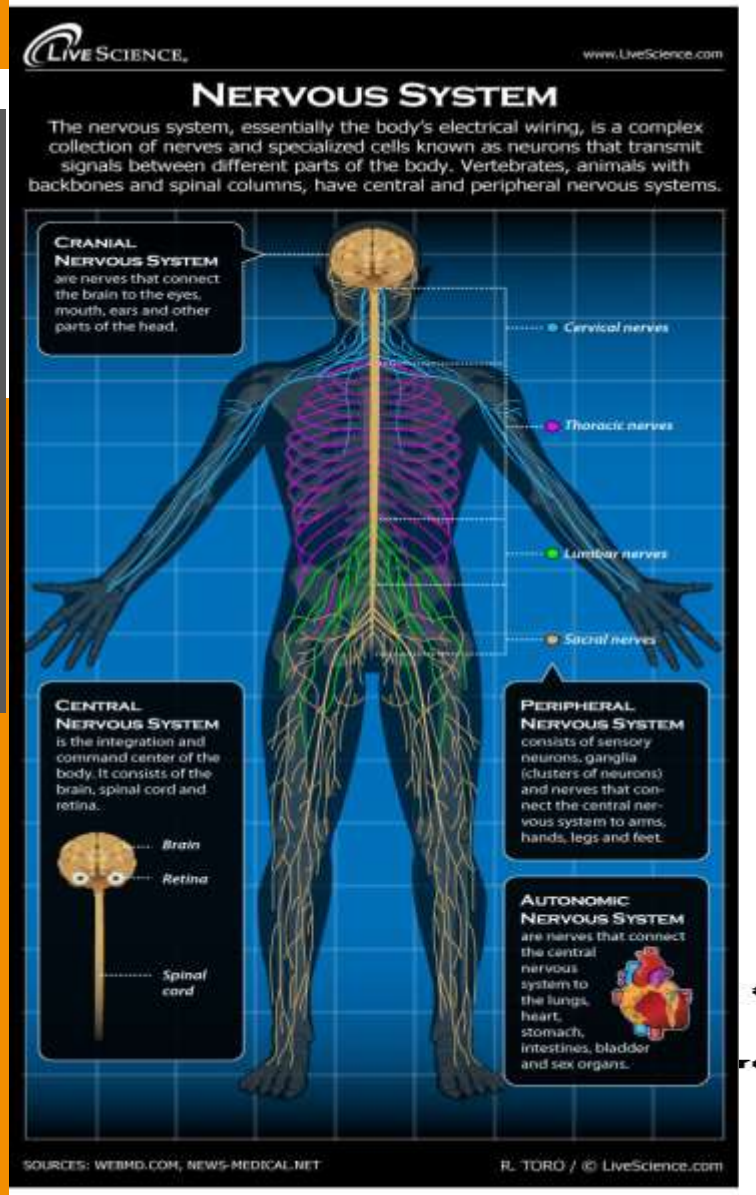
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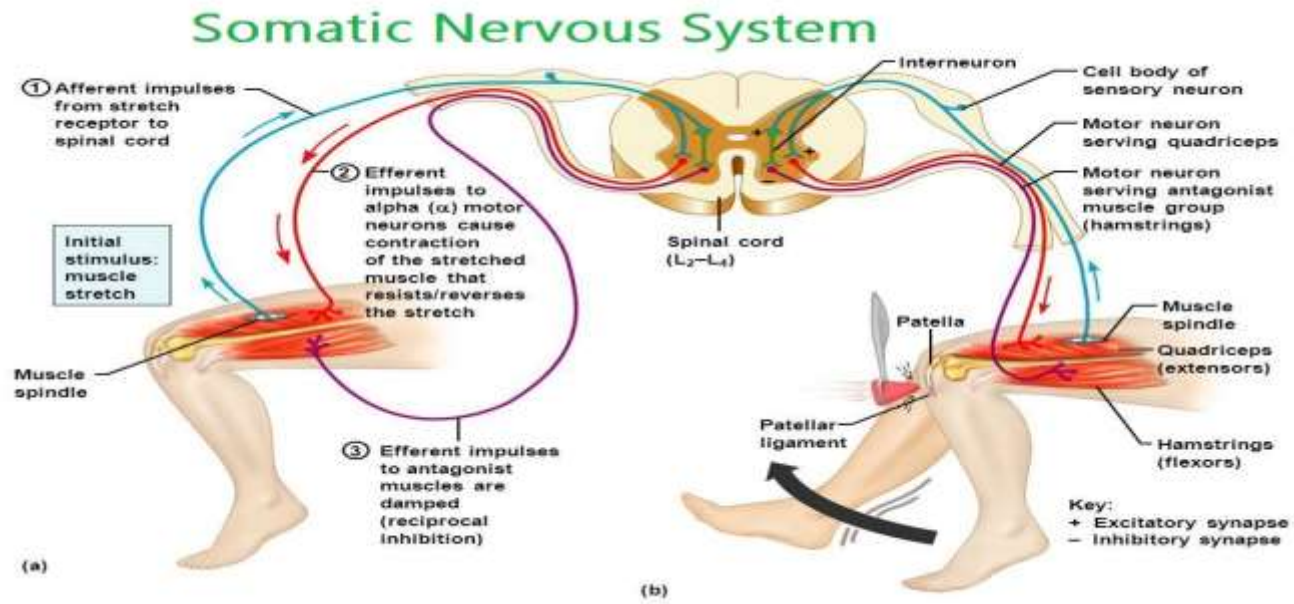
- **Muscular Dystrophy:** Genetic disorder causing muscle weakness.
- **Myasthenia Gravis:** Autoimmune disease leading to fatigue and weakness.
- **Cramps/Strains:** Sudden involuntary contraction of muscles.
- **Tendonitis:** Inflammation of tendons due to overuse.
- **Fibromyalgia:** Chronic widespread muscle pain and fatigue.

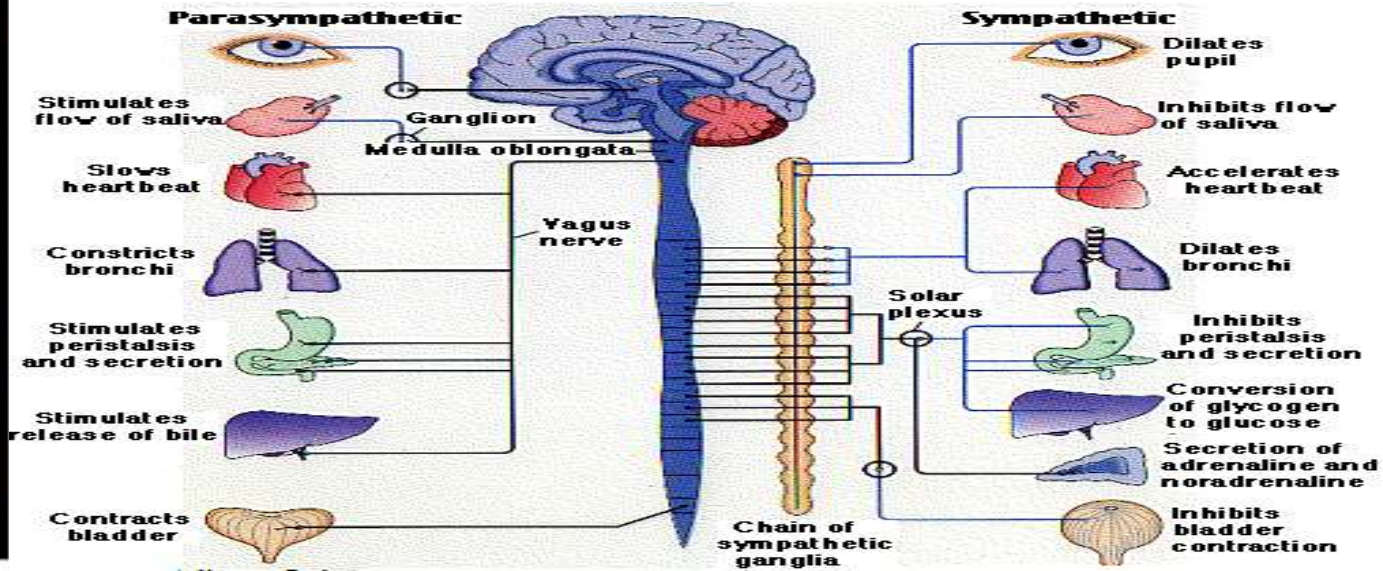
THE NERVOUS SYSTEM

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CNS and PNS

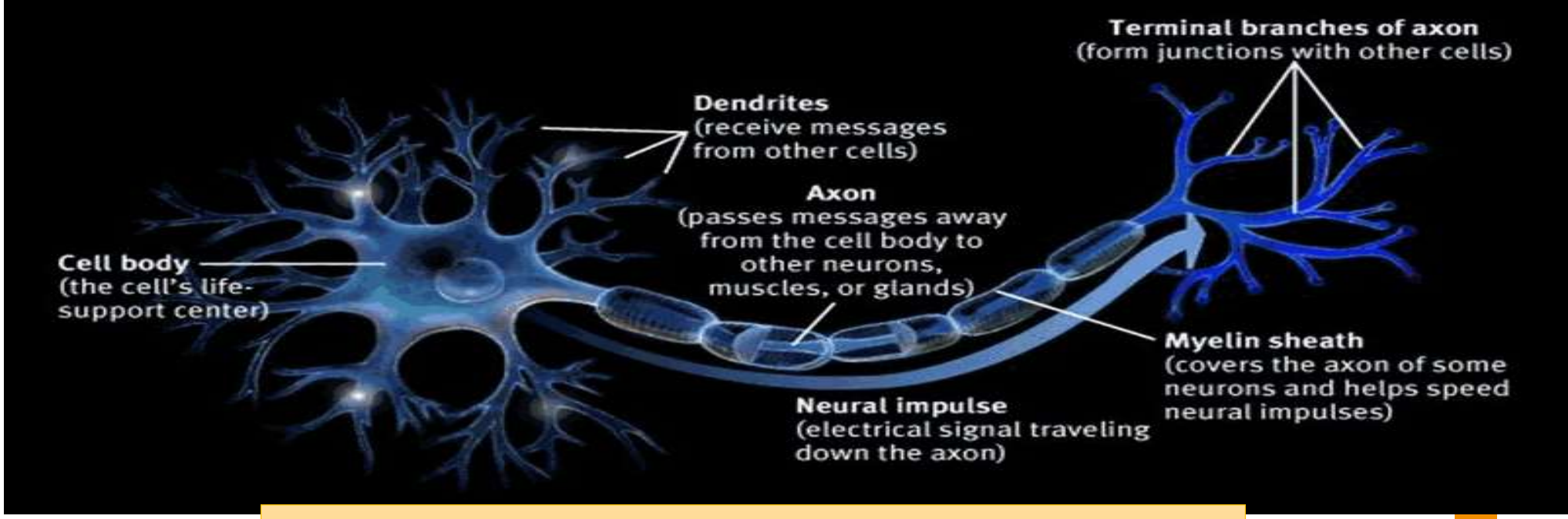
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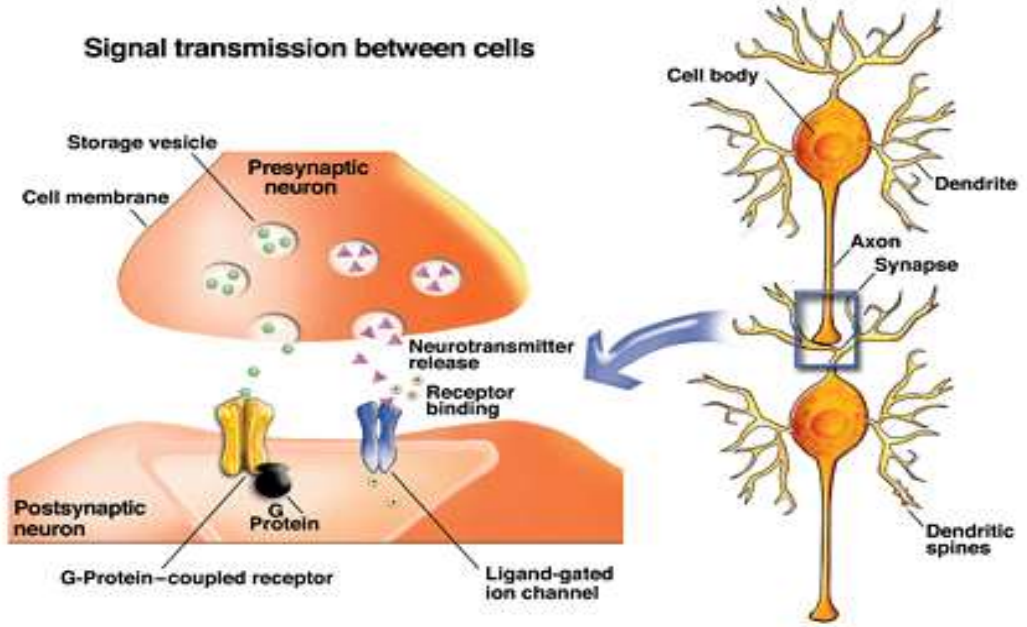
- The Central Nervous System (CNS) and Peripheral Nervous System (PNS) are the two main components of the nervous system in the human body.
- The Central Nervous System consists of the brain and spinal cord and is responsible for receiving, processing, and integrating sensory information and transmitting commands to the rest of the body.
- The brain acts as the command center, receiving and processing sensory inputs and generating motor outputs, while the spinal cord acts as a relay center, transmitting information between the brain and peripheral nerves.
- The Peripheral Nervous System, on the other hand, consists of all the nerves that lie outside the brain and spinal cord.
- It is responsible for transmitting sensory information from the periphery of the body (such as the skin, muscles, and organs) to the CNS, and transmitting commands from the CNS to the periphery.
- The PNS can be further divided into the somatic nervous system and the autonomic nervous system.
- The somatic nervous system controls voluntary movements, while the autonomic nervous system controls involuntary functions such as heart rate, digestion, and respiration.

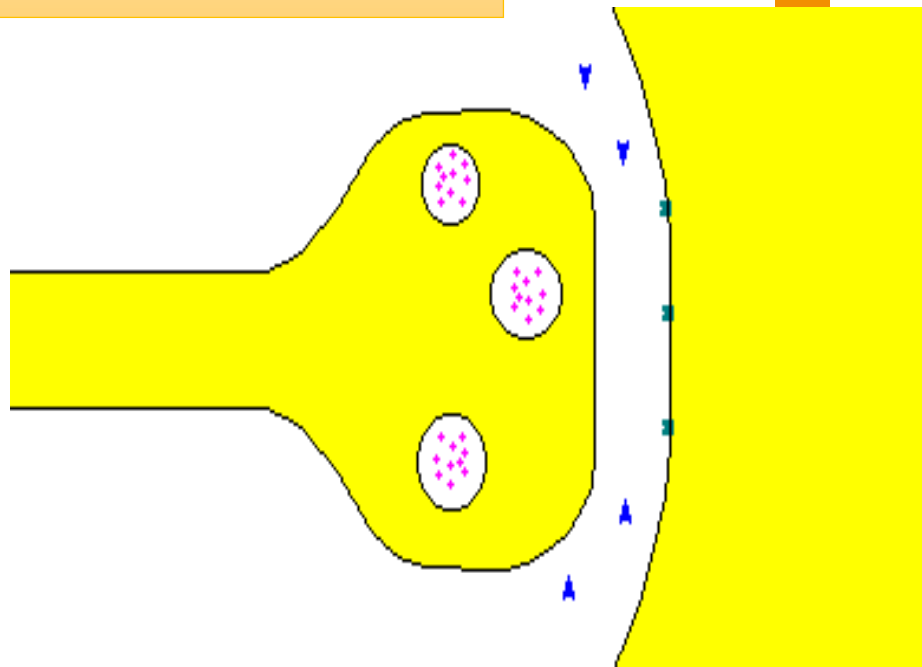
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SIGNAL TRANSMISSION





Functions of the Nervous System

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- **Sensory Input:** Detects changes inside and outside the body via receptors.
- **Integration:** Processes, interprets, and makes decisions.
- **Motor Output:** Responds by activating muscles or glands.
- **Homeostasis:** Maintains stable internal conditions by regulation.
- **Mental Activity:** Enables learning, memory, emotions, language, and reasoning.

Disorders of the Nervous System

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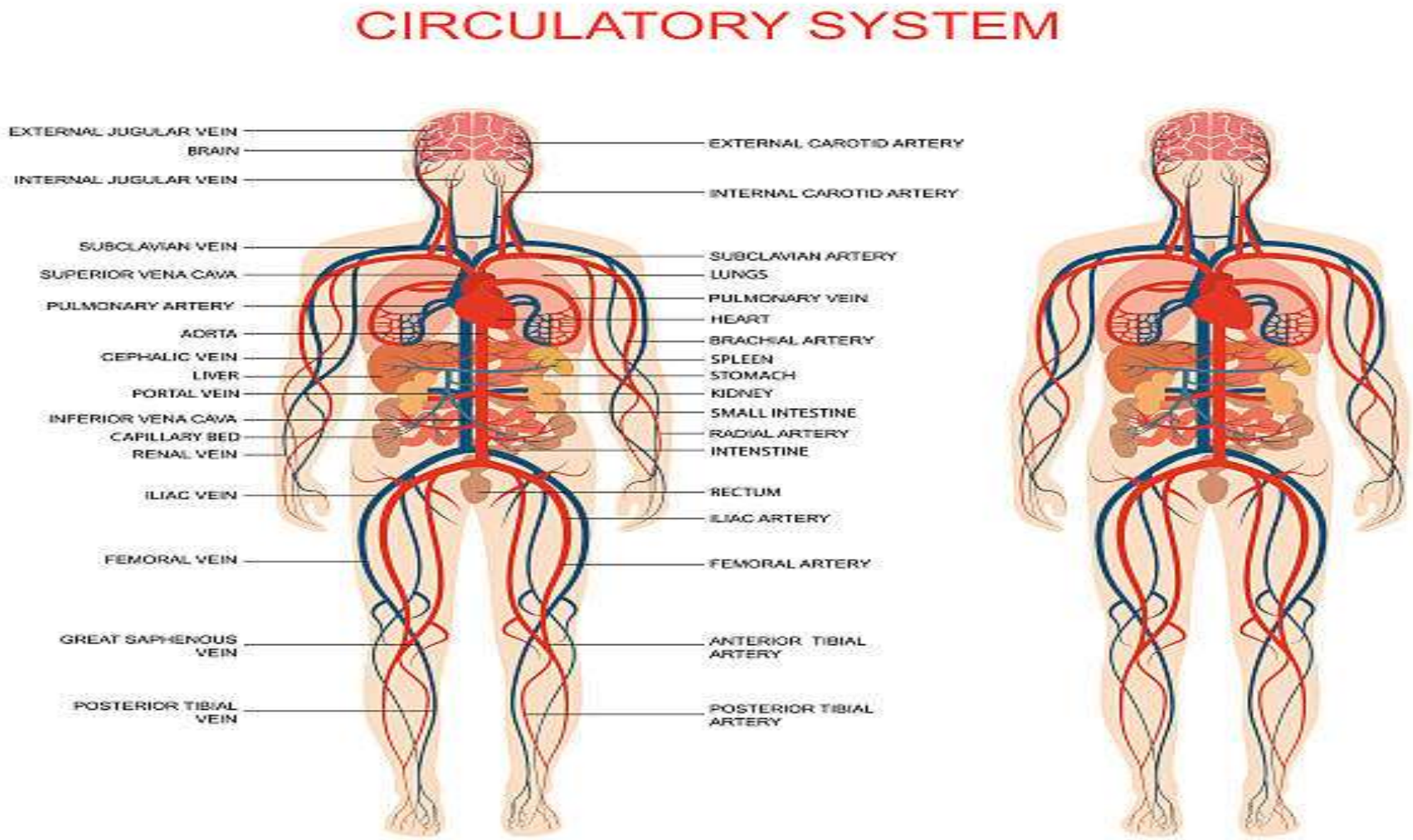
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- **Stroke:** Blockage or rupture of blood vessels in brain.
- **Epilepsy:** Abnormal brain electrical activity causing seizures.
- **Parkinson's Disease:** Degeneration of dopamine-producing neurons causing tremors and rigidity.
- **Alzheimer's Disease:** Progressive memory loss and cognitive decline.
- **Meningitis:** Inflammation of protective membranes (meninges) due to infection.
- **Multiple Sclerosis (MS):** Autoimmune destruction of myelin sheath.

CIRCULATORY SYSTEM

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Components of the Circulatory System

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- **1. Heart**
- Muscular, cone-shaped organ located in the thoracic cavity.
- Size: About the size of a fist, weight ~250–300 g.
- Functions as a **pump** to circulate blood.
- **Structure of the Heart:**
- **Chambers:**
  - 2 atria (upper chambers) – receive blood.
  - 2 ventricles (lower chambers) – pump blood out.
- **Valves:** Ensure unidirectional blood flow (tricuspid, bicuspid/mitral, pulmonary, aortic).
- **Septum:** Divides right and left sides of the heart.
- **Circulation Pathways:**
- **Pulmonary Circulation:** Right ventricle → lungs → left atrium (oxygenates blood).
- **Systemic Circulation:** Left ventricle → body → right atrium (supplies oxygen/nutrients to tissues).

Components of the Circulatory System

- **2. Blood Vessels**
  - **Arteries:** Carry blood **away** from heart (mostly oxygen-rich, except pulmonary artery). Thick, muscular walls.
  - **Veins:** Carry blood **toward** heart (mostly oxygen-poor, except pulmonary vein). Have valves to prevent backflow.
  - **Capillaries:** Tiny thin-walled vessels where exchange of gases, nutrients, and wastes occurs.
- **Blood**
  - Total volume: ~5 liters in adults.
  - **Components:**
    - **Plasma (55%)** – fluid part; carries nutrients, hormones, waste.
    - **Red Blood Cells (RBCs):** Carry oxygen using hemoglobin.
    - **White Blood Cells (WBCs):** Defend against infections.
    - **Platelets:** Help in clotting to prevent blood loss.

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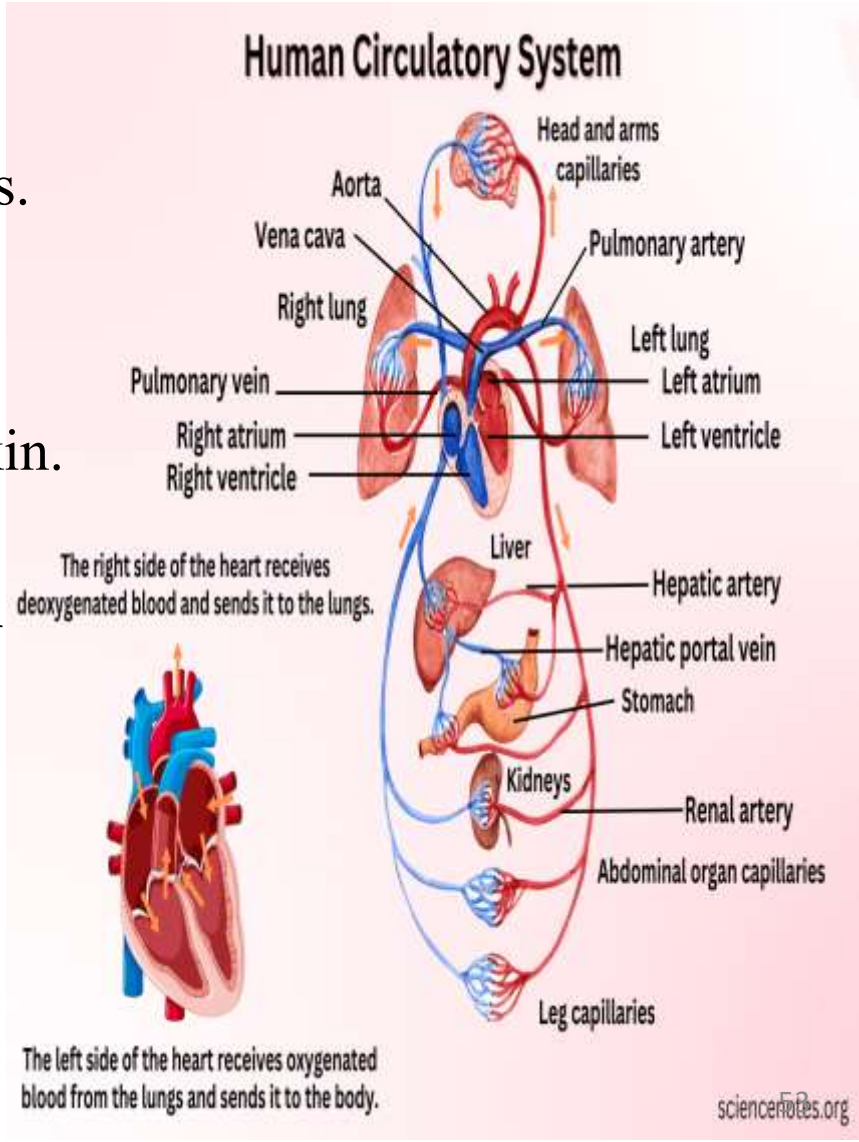
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Functions of the Circulatory System

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- **Transport:**
  - Oxygen from lungs → body tissues.
  - Carbon dioxide from tissues → lungs.
  - Nutrients from digestive system → cells.
  - Hormones → target organs.
  - Waste products → kidneys, lungs, skin.
- **Regulation:** Maintains body temperature, pH balance, and fluid balance.
- **Protection:**
  - WBCs fight infections.
  - Platelets prevent blood loss.
  - Circulating antibodies provide immunity.



Disorders of the Circulatory System

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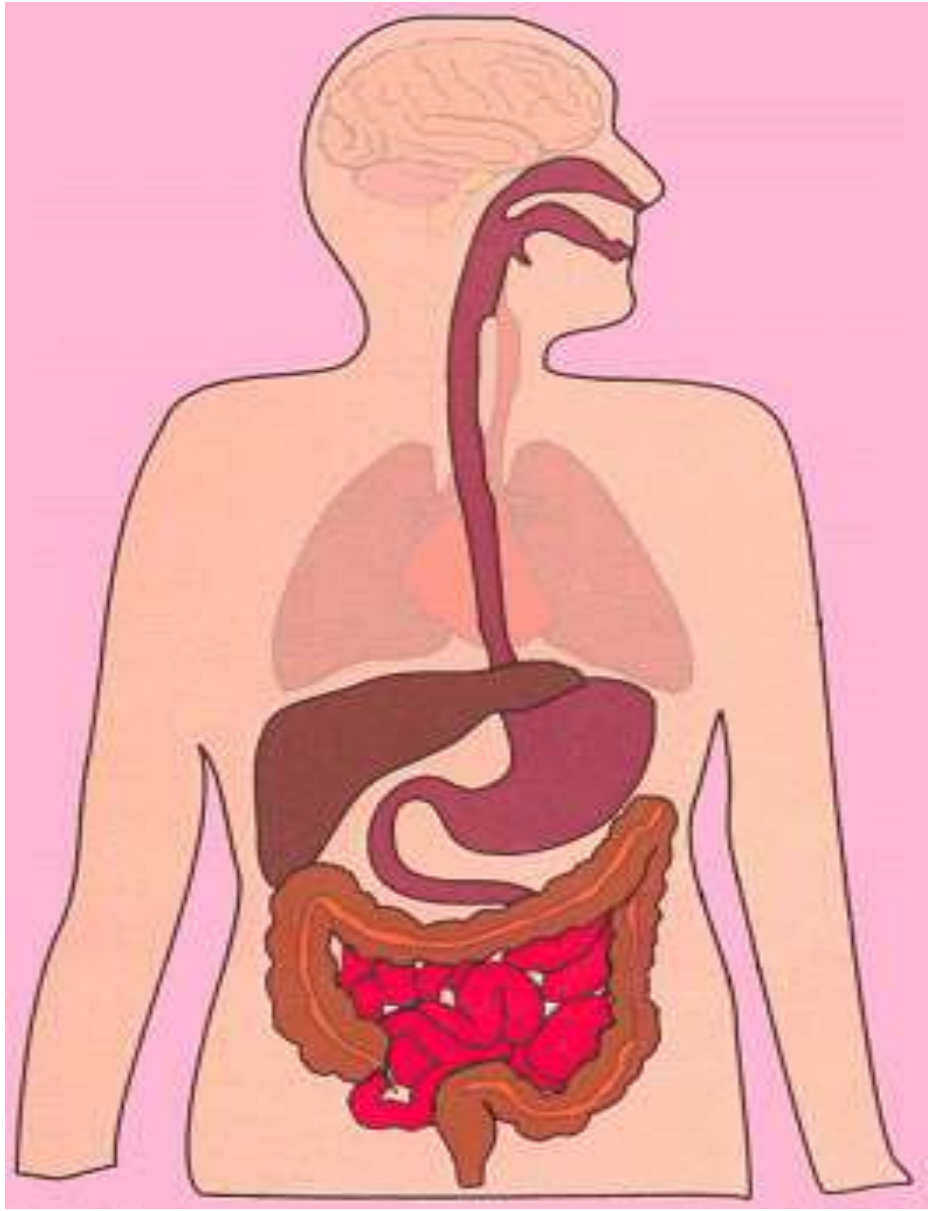
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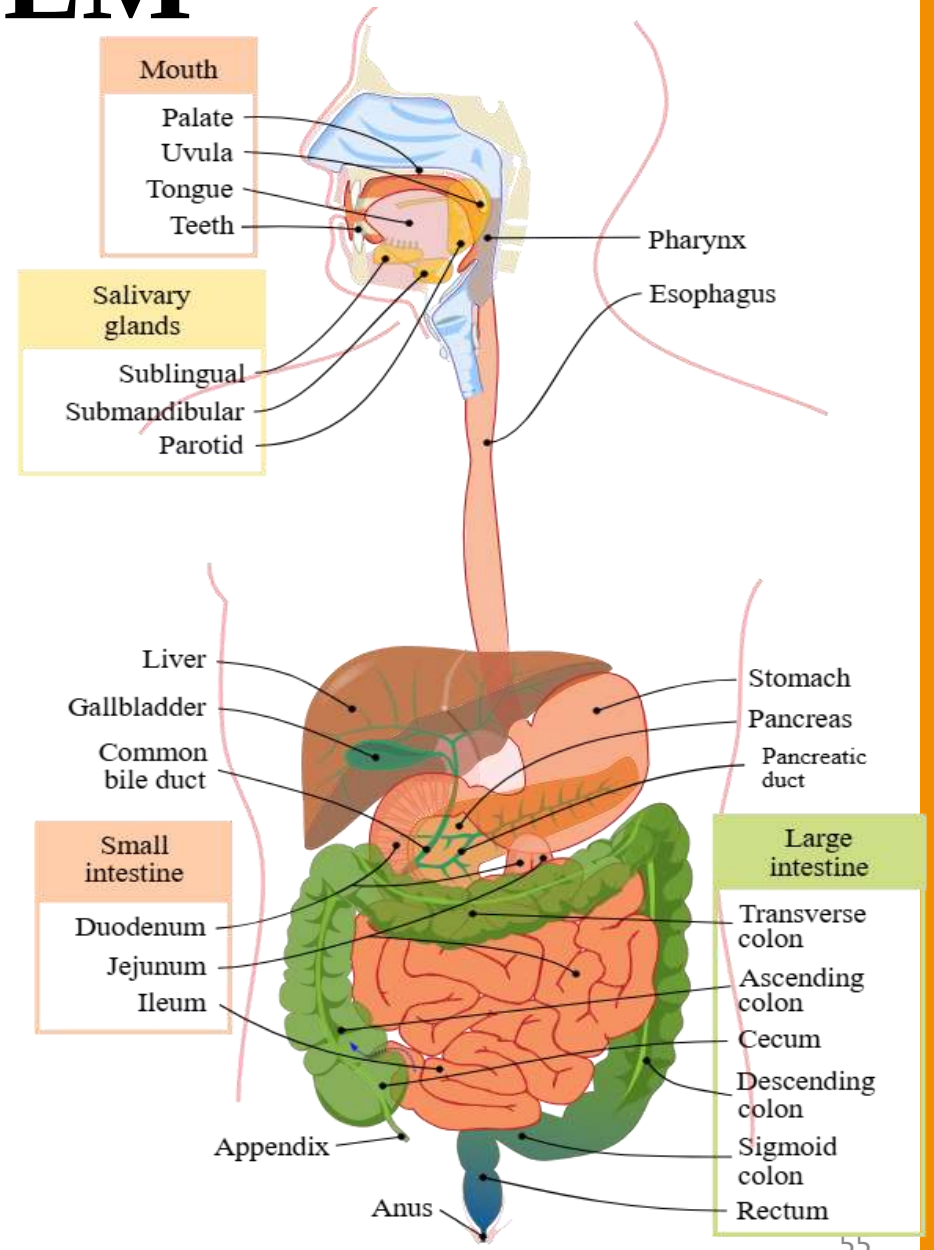
- **Hypertension (High BP):** Persistent elevation of blood pressure.
- **Atherosclerosis:** Hardening/narrowing of arteries due to cholesterol deposits.
- **Heart Attack (Myocardial Infarction):** Blockage of coronary artery → death of heart tissue.
- **Stroke:** Blockage or rupture of brain blood vessels.
- **Anemia:** Low hemoglobin or RBC count.
- **Leukemia:** Cancer of WBCs.
- **Varicose Veins:** Swollen veins due to valve failure.

DIGESTIVE SYSTEM

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Major Parts of the Digestive System

1. Alimentary Canal (GI Tract)

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Mouth	Pharynx & Esophagus	Stomach	Small Intestine (duodenum, jejunum, ileum)	Large Intestine (Colon)	Rectum & Anus
First site of digestion. Teeth break food into small pieces (mechanical digestion). Saliva (from salivary glands) contains **amylase enzyme** which begins starch digestion (chemical digestion). Tongue helps in swallowing.	Food passes through pharynx and enters the esophagus. **Peristalsis** (wave-like muscular contractions) moves food toward stomach.	Muscular sac that stores food temporarily. Gastric glands secrete **hydrochloric acid (HCl)** and **pepsin enzyme** to digest proteins. Food is converted into semi-liquid substance called **chyme**.	Main site of digestion and absorption. **Duodenum:** Receives bile (from liver) and pancreatic juices (from pancreas) for digestion of fats, proteins, and carbohydrates. **Jejunum & Ileum:** Absorb nutrients into the bloodstream through villi and microvilli.	Absorbs water, electrolytes, and vitamins (produced by gut bacteria). Compacts undigested material into feces.	Stores feces until excretion. Anus allows controlled elimination (defecation).

The larynx and esophagus work together during the swallowing phase of digestion to safely guide chewed food from the mouth to the stomach without letting anything enter the airway

2. **Accessory Digestive Organs**

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Liver – Produces **bile**, which helps emulsify fats.

Gallbladder – Stores and releases bile into duodenum.

Pancreas – Secretes digestive enzymes (amylase, lipase, proteases) and bicarbonate to neutralize stomach acid.

Salivary Glands – Produce saliva to aid in chewing and swallowing.

Functions of the Digestive System

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- **Ingestion** – Taking in food and drink.
- **Propulsion** – Moving food along (swallowing and peristalsis).
- **Mechanical Digestion** – Chewing, churning in stomach, segmentation in intestines.
- **Chemical Digestion** – Enzymatic breakdown of carbohydrates, proteins, and fats.
- **Absorption** – Transfer of nutrients into blood and lymph.
- **Defecation** – Elimination of indigestible substances as feces.

Enzymes and Secretions in Digestion

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- **Mouth:** Salivary amylase (starch → maltose).
- **Stomach:** Pepsin (proteins → peptides), HCl (kills microbes, activates enzymes).
- **Pancreas:**
  - Amylase (carbs → maltose),
  - Lipase (fats → fatty acids + glycerol),
  - Trypsin & Chymotrypsin (proteins → peptides).
- **Small Intestine:** Peptidases, maltase, lactase, sucrase (for final breakdown).
- **Bile (from liver):** Emulsifies fats (mechanical digestion, not enzymatic).

Disorders of the Digestive System

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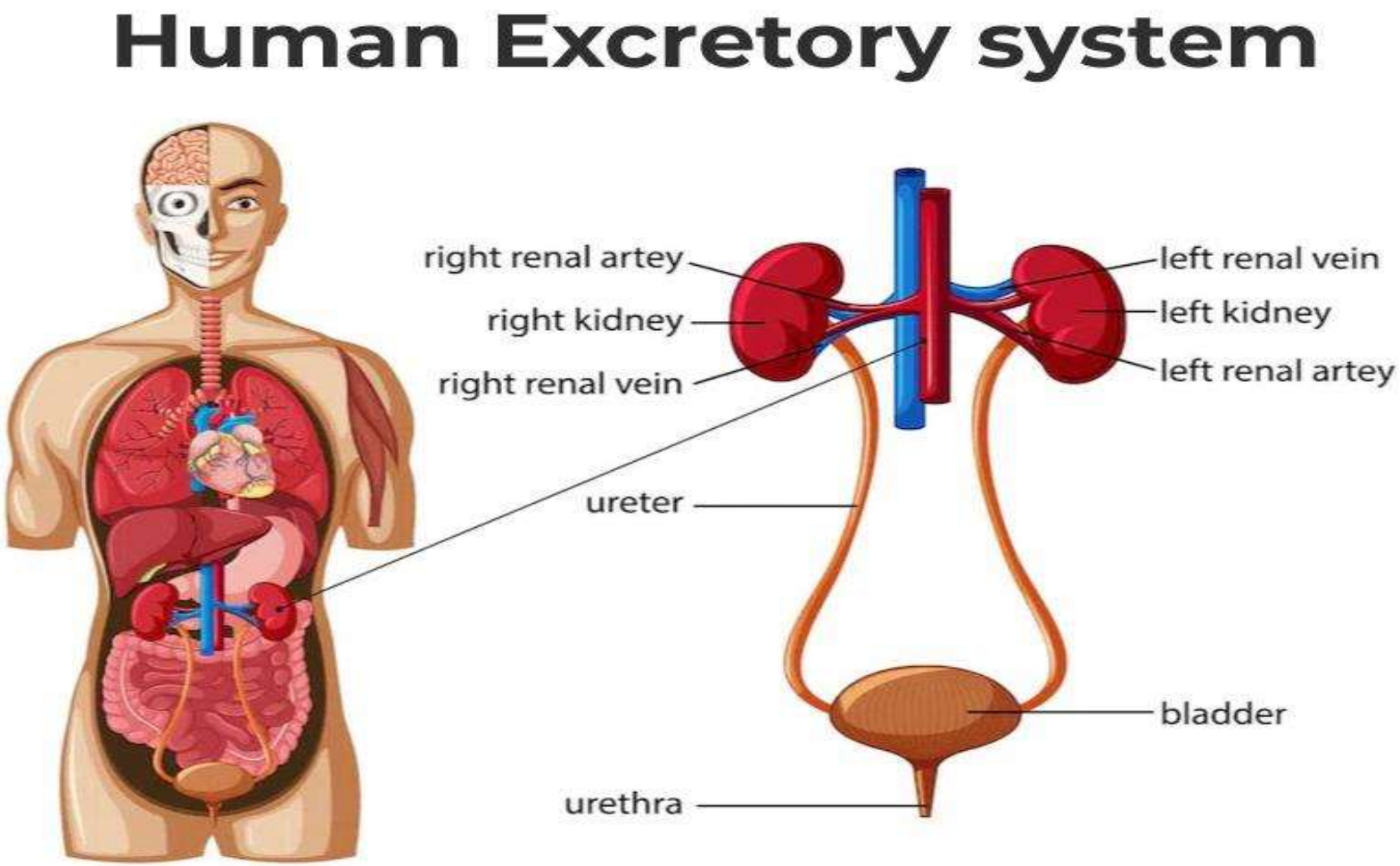
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- **Gastritis:** Inflammation of stomach lining.
- **Ulcers:** Erosion of stomach/duodenal wall by acid.
- **Diarrhea:** Rapid movement of intestinal contents → watery stools.
- **Constipation:** Slow movement of feces → hard stools.
- **Gallstones:** Hardened deposits in gallbladder.
- **Hepatitis:** Inflammation of liver (caused by viral infection).
- **GERD (Acid Reflux):** Acid moves back into esophagus.

EXCRETORY SYSTEM

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Main Organs of the Human Excretory System

- **1. Kidneys**
- Bean-shaped organs, ~10–12 cm long, located in the abdominal cavity.
- Each kidney has about **1 million nephrons** (functional units).
  - **Structure of Nephron:**
    - **Bowman's capsule**: Receives blood filtrate from glomerulus.
    - **Glomerulus**: Capillary network where filtration occurs.
    - **Proximal convoluted tubule (PCT)**: Reabsorbs water, ions, glucose, amino acids.
    - **Loop of Henle**: Maintains water and salt balance (concentrates urine).
    - **Distal convoluted tubule (DCT)**: Selective secretion and reabsorption.
    - **Collecting duct**: Final urine formation and transport.
- **Functions of Kidneys:**
- Remove nitrogenous wastes (**urea, uric acid, creatinine**).
- Regulate water balance (osmoregulation).
- Regulate electrolyte balance (Na^+, K^+, Ca^{2+}).
- Maintain blood pressure (via renin enzyme).
- Produce hormones (**erythropoietin for RBC production, calcitriol for calcium absorption**).

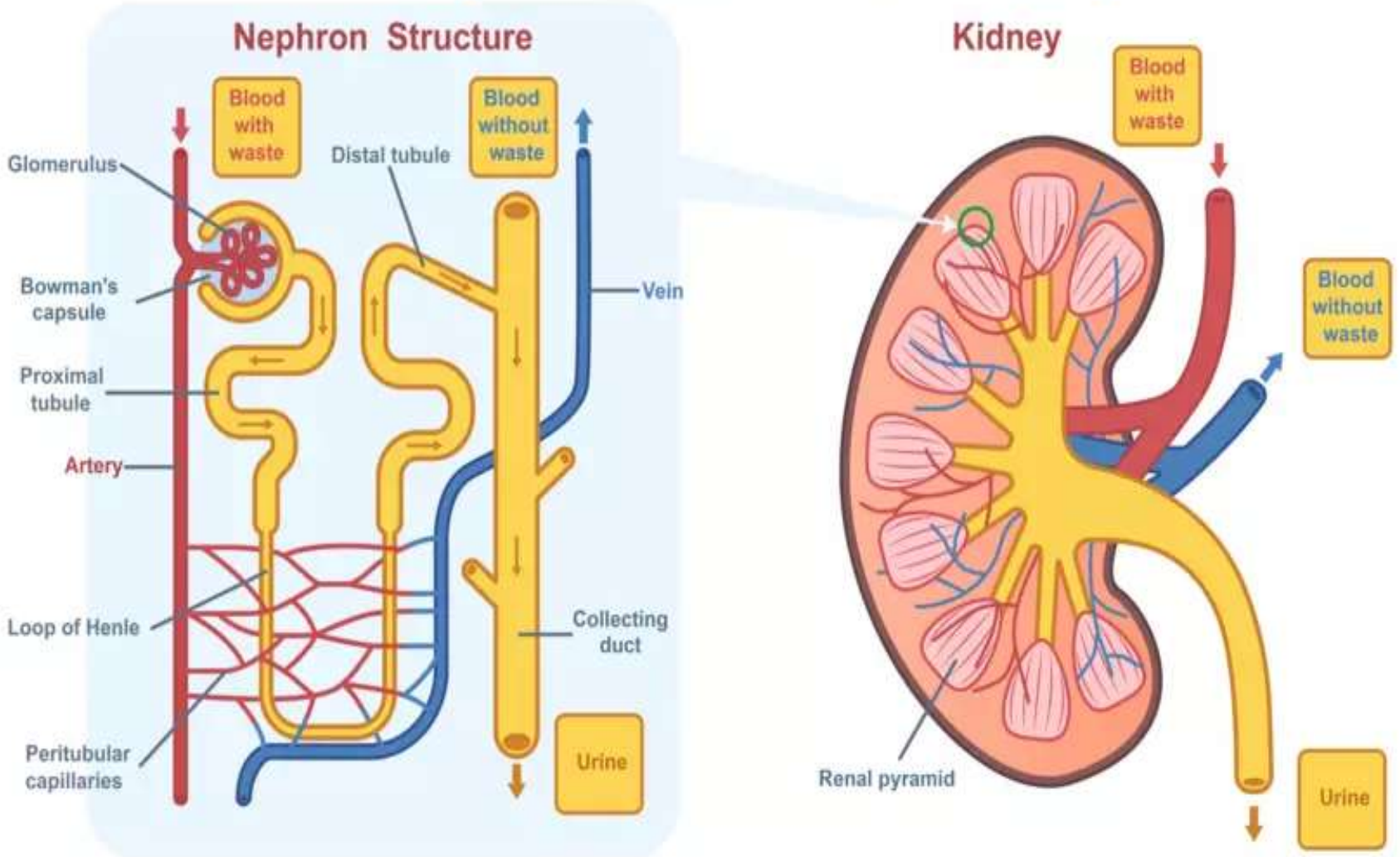
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Nephron Anatomy



Main Organs of the Human Excretory System

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- **2. Ureters**
- Two narrow muscular tubes (~25–30 cm).
- Carry urine from kidneys to urinary bladder by **peristalsis**.
- **3. Urinary Bladder**
- Muscular sac for temporary urine storage (~400–600 ml capacity).
- Lined with transitional epithelium to allow stretching.
- **4. Urethra**
- Tube that carries urine out of the body.
- Controlled by sphincter muscles.
  - In males – longer (~20 cm), common passage for urine & semen.
  - In females – shorter (~4 cm), only urine passage.

Functions of the Excretory System

- Removal of metabolic wastes.
- Regulation of water and salt balance.
- Maintenance of acid–base balance (blood pH).
- Regulation of blood pressure.
- Hormone production for RBC formation & calcium regulation.

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Disorders of the Excretory System

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- **Kidney Stones:** Hard deposits of calcium/uric acid blocking urinary tract.
- **Urinary Tract Infections (UTI):** Bacterial infection of bladder/kidneys.
- **Glomerulonephritis:** Inflammation of glomeruli, affecting filtration.
- **Renal Failure:** Loss of kidney function, requiring dialysis.
- **Proteinuria:** Presence of protein in urine (sign of kidney damage).

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Metabolism and control of carbohydrates, lipids and proteins.

Metabolism of Carbohydrates

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- **Metabolism of carbohydrates** refers to the chemical processes that break down carbohydrates to produce energy, as well as those that synthesize necessary carbohydrates for the body. The primary pathway for carbohydrate metabolism is **glycolysis**, where glucose is broken down into pyruvate to generate ATP (energy).

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Breakdown and Energy Production

•**Glycolysis**: Splits one glucose molecule in the cell cytoplasm into two pyruvate molecules, producing a small amount of ATP.
•**Citric Acid Cycle (Krebs Cycle)**: Processes pyruvate inside mitochondria aerobically to create high-energy electron carriers (NADH and $FADH_2$).
•**Electron Transport Chain**: Uses electrons from the Krebs cycle to generate large amounts of ATP via oxidative phosphorylation.

Storage and Synthesis

•**Glycogenesis**: Converts extra glucose into glycogen for storage in the liver and skeletal muscles.
•**Glycogenolysis**: Breaks stored glycogen back down into glucose when the body needs fast energy during fasting.
•**Gluconeogenesis**: Builds fresh glucose from non-sugar sources like fats or amino acids during low-carbohydrate intake or starvation.

Glycolysis

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- **Definition**: Glycolysis is the pathway that breaks down one molecule of **glucose (6 carbons)** into **two molecules of pyruvate (3 carbons each)**.
- **Location**: Cytoplasm of the cell.
- **Process**:
  - Glucose → Glucose-6-phosphate → Fructose-1,6-bisphosphate → Glyceraldehyde-3-phosphate → Pyruvate
- **Energy Yield**:
  - Net gain of **2 ATP** molecules and **2 NADH** molecules per glucose.
- **Importance**:
  - Provides energy quickly.
  - Occurs in both **aerobic and anaerobic** conditions.
  - In anaerobic conditions (e.g., muscle during exercise), pyruvate is converted into **lactate**.

Gluconeogenesis

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- **Definition**: It is the process of **synthesizing glucose** from **non-carbohydrate sources**, such as **amino acids**, **glycerol**, and **lactate**.
- **Location**: Mainly in the **liver** and to some extent in the **kidneys**.
- **When it occurs**:
  - During fasting or starvation.
  - When carbohydrate intake is low.
- **Purpose**:
  - Maintains blood glucose levels for **brain and red blood cells**, which rely on glucose exclusively for energy.
- **Key substrates**:
  - Lactate (from muscle)
  - Glycerol (from fat)
  - Alanine and other amino acids (from muscle protein)

Glycogenesis

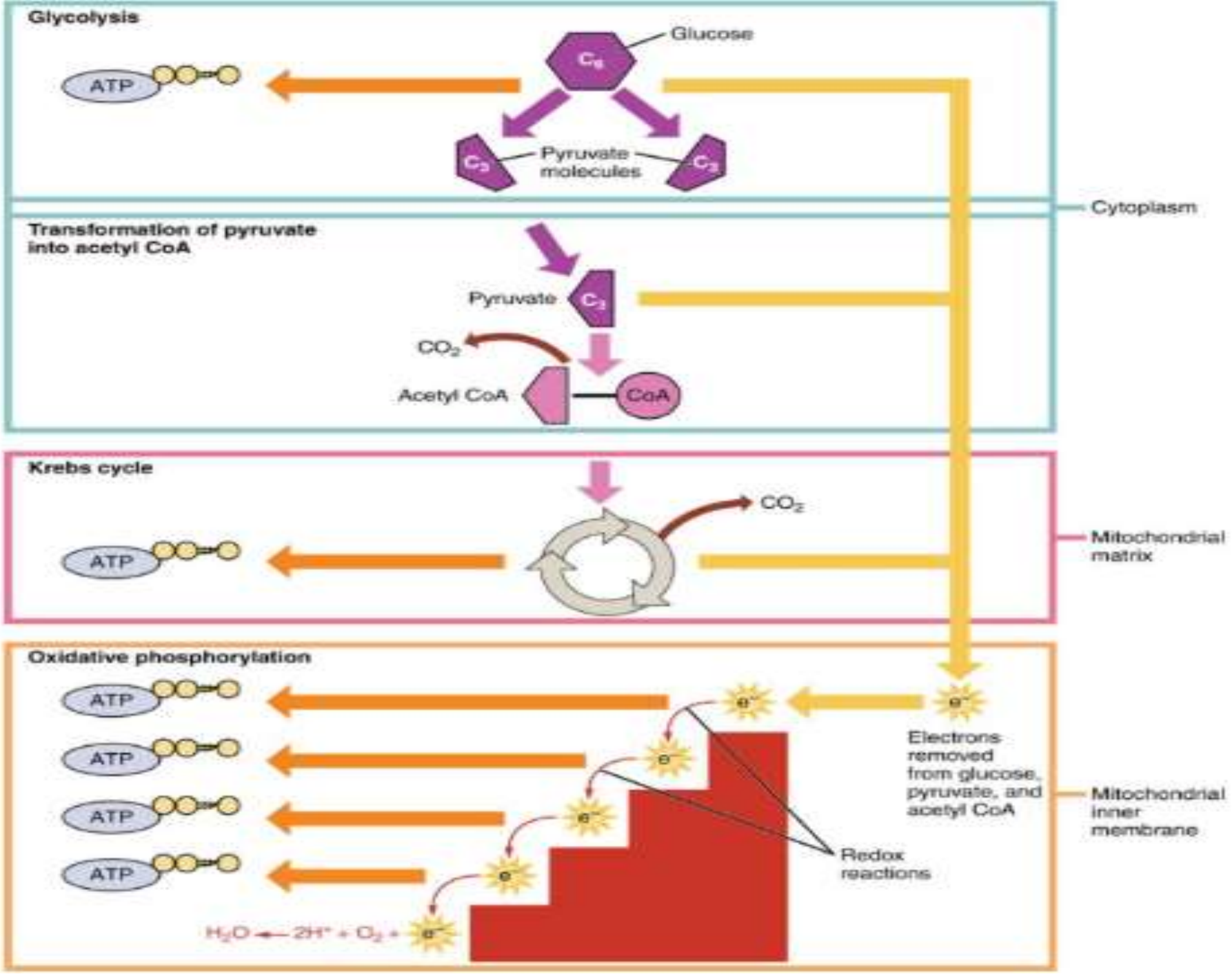
- **Definition**: It is the process of converting **excess glucose** into **glycogen** for storage.
- **Location**: Liver and skeletal muscles.
- **Process**:
  - Glucose → Glucose-6-phosphate → Glucose-1-phosphate → UDP-glucose → Glycogen
- **Function**:
  - In the liver: Maintains **blood glucose levels**.
  - In muscles: Provides **energy** for muscle contraction.

Glycogenolysis

- **Definition**: It is the breakdown of **glycogen** back into **glucose-1-phosphate**, which is then converted to **glucose-6-phosphate**.
- **Location**: Liver and muscle.
- **Purpose**:
  - In the **liver**: Releases free glucose into the bloodstream to maintain blood glucose levels.
  - In **muscles**: Provides glucose-6-phosphate for **ATP production** during activity.

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Lipid metabolism

Lipid metabolism is the synthesis and degradation of lipids in cells, involving the breakdown and storage of fats for energy and the synthesis of structural and functional lipids, such as those involved in the construction of cell membranes. In animals, these fats are obtained from food and are synthesized by the liver

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Metabolism and control of lipids

- **Site of Digestion:** Begins in the small intestine (minimal digestion in the mouth/stomach).
- **Key Enzymes:** Bile salts (from liver/gallbladder): Emulsify fats to increase surface area.
- **Pancreatic lipase:** Breaks triglycerides into free fatty acids and monoglycerides.
- **Absorption:** Products of digestion enter intestinal cells.
- Reassembled into triglycerides and packaged into chylomicrons (lipoproteins).
- Chylomicrons enter the lymphatic system, then bloodstream.

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Breakdown of Fats (Catabolism)

- Lipolysis: Stored fats (triglycerides) break apart into fatty acids and glycerol when the body needs extra fuel.
- Beta-oxidation: Fatty acids turn into acetyl-CoA inside cell mitochondria to create energy.
- Ketogenesis: The liver changes extra acetyl-CoA into ketone bodies to feed the brain and muscles during fasting

Making of Fats (Anabolism)

- Lipogenesis: The body builds new triglycerides from extra food calories and carbohydrates to store as body fat.
- Cholesterol synthesis: The liver uses acetyl-CoA to make cholesterol for cell walls and hormones.

Metabolism and control of Lipids

- **Lipid Catabolism (Breakdown)**
- A**. Lipolysis**
  - Triglycerides in adipose tissue are broken into: Glycerol (enters glycolysis or gluconeogenesis)
  - Free fatty acids (used for energy)
- **B. Beta-Oxidation**
  - Occurs in mitochondria. Fatty acids are broken down into acetyl-CoA.
  - Acetyl-CoA enters the TCA cycle to produce ATP.
- **C. Ketogenesis (in starvation or diabetes)**
  - When carbohydrates are low, excess acetyl-CoA in liver is converted into:
  - Ketone bodies (acetoacetate, β-hydroxybutyrate) Used by brain and muscles as alternative energy source.
  - Energy Yield from Lipids
  - Fats are high-energy molecules: 1 gram of fat yields 9 kcal, compared to 4 kcal from carbs/proteins.
  - Complete oxidation of 1 molecule of palmitic acid (16-carbon fatty acid) produces 129 ATP.

Metabolism and control of Proteins

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- Digestion begins in the stomach with the enzyme pepsin, which breaks proteins into smaller peptides.
- In the small intestine, enzymes like trypsin, chymotrypsin, and peptidases further break down peptides into amino acids.
- These amino acids are then absorbed into the bloodstream through the intestinal wall and transported to the liver.
- Once absorbed, amino acids form an **amino acid pool** in the body, which is used for:
- **Protein synthesis** (muscle, enzymes, hormones)
- **Energy production** (when needed)
- **Synthesis of nitrogenous compounds** (e.g., neurotransmitters, nucleotides)
- Amino acids are **not stored**, so excess is broken down.

Metabolism and control of Proteins

- **Protein Synthesis (Anabolism)**
  - Occurs in **ribosomes** using genetic instructions from DNA (via mRNA).
  - **tRNA** brings specific amino acids to the ribosome where they are joined by **peptide bonds** to form proteins.
  - **Tissue repair**, **enzyme synthesis**, and **hormone production** depend on this process.
- **Protein Breakdown (Catabolism)**
  - When the body requires energy or is under stress (e.g., fasting, illness), proteins are broken down.
  - **Steps:**
  - **Deamination**:
    - Removal of the amino group (—NH_2) from amino acids.
    - Occurs mainly in the **liver**.
    - Produces **ammonia (NH_3)**, which is toxic.
  - **Urea Cycle**:
    - Ammonia is converted into **urea** in the liver and excreted via the kidneys.
  - **Carbon Skeleton Utilization**:
    - The remaining carbon skeleton is converted into:
      - **Glucose** (via gluconeogenesis)
      - **Ketone bodies**
      - **ATP** (via the TCA cycle)

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Food Constituent

I. Major Food Components

1.Water
2. Protein
3. Carbohydrates
4. Fats

II Minor Food Components

1. Food Phenolics
2. Natural emulsifiers
3. Organic acids
4. Oxidants and antioxidants
5. Enzymes
6. Pigments and colors
7. Flavors
8. Vitamins
9. Minerals
10. Analogs and new ingredients

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Vitamins and minerals

Vitamins and minerals are organic compounds that our bodies use in very small amounts for a variety of metabolic processes. Basically, they keep us healthy and help our bodies to function.

There are 13 essential vitamins divided into two main groups:

- Fat-soluble vitamins (A, D, E, K)
- Water-soluble vitamins (the eight B-complex vitamins and vitamin C)

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Vitamin A

- Makes the immune system work effectively so it can fight disease and infections
- keeps our skin healthy
- supports reproduction and growth
- helps with vision.

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Vitamin B

The 8 types of vitamin B are:

thiamin (B1)

riboflavin (B2)

niacin (B3)

pantothenic acid (B5)

pyridoxine (B6)

biotin (B7)

folate or ‘folic acid’ when included in supplements (B9)

cyanocobalamin (B12).

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Vitamin C

Vitamin C (ascorbic acid) is important for many metabolic processes, including:

Collagen formation – collagen is used in different ways throughout the body. Its primary role is to strengthen the skin, blood vessels and bone. The body also relies on collagen to heal wounds.

Antioxidant function– the metabolism of oxygen within the body releases molecular compounds called ‘free radicals’, which damage cell membranes. Antioxidants are substances that destroy free radicals, and vitamin C is a powerful antioxidant.

Iron absorption – the process of iron absorption is aided by vitamin C, particularly non-haem iron (found in plant foods such as beans and lentils).

Infection fighting – the immune system, particularly cells called lymphocytes, requires vitamin C for proper functioning.

Other roles – vitamin C is used to produce other important substances in the body such as brain chemicals (neurotransmitters).

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Vitamin D

- Vitamin D is important for strong bones, muscles and overall health. Ultraviolet (UV) radiation from the sun is necessary to produce vitamin D in the skin and is the best natural source of vitamin D.
- Regular physical activity also assists with the body's production of vitamin D.
- The body can only absorb small amounts of Vitamin D.

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Vitamin E

- Vitamin E is best obtained from a healthy diet that contains plenty of fresh minimally processed foods. Vitamin E is also vulnerable to heat (especially cooking methods such as deep frying.

Vitamin K

- healthy bones
- blood clotting and wound healing
- newborn babies to prevent a serious bleeding condition called haemorrhagic disease of the newborn (HDN).

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Calcium

- Calcium is vital to keep our bones strong and healthy. If you don’t get enough calcium, your bones will eventually become weak and brittle and can lead to conditions like osteoporosis. Calcium helps:

- strengthen bones and teeth
- regulate muscle and heart function
- blood clotting
- transmission of nervous system messages
- enzyme function.
- Food sources o

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Iodine

- Iodine is essential to make thyroid hormones. These hormones control your metabolic rate (the rate your body uses energy when it is resting). They also help your brain and body grow and develop.

Iron

- Iron is an important mineral that is involved in various bodily functions, including the transport of oxygen in the blood the provision of energy to cells. It also vital to help our immune system function effectively to fight infection.

Zinc

- Zinc is an important mineral involved in various bodily functions – growth and development as well as immune function.
- Zinc also helps to produce the active form of vitamin A and transports it around the body.

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Magnesium

- Magnesium is important due to its many functions in the body – including maintaining bone health and using glucose for energy.
- Magnesium also supports immune function and helps regulate blood pressure and lung function.
- Magnesium is the essential mineral known for its soothing, calming effect on the body and mind, which helps promote good, restful sleep.

Potassium

- Potassium is important for the nerves, muscles and heart to work properly. It also helps lower

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Sodium

- A small amount of sodium is important for good health as it helps to maintain the correct volume of circulating blood and tissue fluids in the body.

- Most of us are consuming far more sodium than we need. In fact, many Australians are consuming almost double the amount required.

- Too much sodium can lead to high blood pressure (hypertension) and other health conditions.